

Chapter 7. Intermittent Mandatory Ventilation

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Intermittent Mandatory Ventilation: Introduction

Intermittent mandatory ventilation (IMV) allows the patient to breathe spontaneously between machine-cycled or mandatory breaths. This concept originated in 1955 with an unnamed ventilator designed by Engstrom.^{1,2} In the early 1970s, Kirby et al^{3,4} introduced IMV as a means of ventilator support of infants with respiratory distress syndrome. In 1973, Downs et al⁵ were the first to propose IMV as a method to facilitate discontinuation from mechanical ventilation in adults. Those investigators^{6,7} also pioneered IMV use as a primary means of ventilator support during acute respiratory failure. Subsequently, breath-delivery design has been modified. Mandatory breaths initially delivered regardless of respiratory timing are synchronized with the patient's inspiratory effort.^{8,9} This mode of ventilation has been termed *intermittent demand ventilation*,⁸ *intermittent assisted ventilation*,⁹ and *synchronous intermittent mandatory ventilation* (SIMV). SIMV is an established partial mechanical ventilation mode in critically ill patients, both adult¹⁰ and neonate, worldwide.¹¹ Currently, however, SIMV application in adults has declined except in North America¹² and Australia–New Zealand,¹³ whereas in neonates, SIMV application remains prevalent.¹⁴ This chapter uses the terms *IMV* and *SIMV* interchangeably unless specifically indicated for clarification.

Basic Principles

Description

IMV is a means of ventilator support in which a preset number of positive-pressure (mandatory) breaths are delivered while the patient breathes spontaneously between the mandatory breaths. The mandatory breaths can be in the form of a preset volume (flow-limited, volume-cycled), pressure (pressure-limited, time-cycled),¹⁵ or a combination of pressure and volume (dual control).¹⁶ In principle, IMV is similar to controlled mechanical ventilation (CMV), in which the patient receives a predetermined number of mandatory machine-triggered breaths independent of spontaneous breathing effort. Likewise, SIMV is similar to assist-control ventilation (ACV), in which mandatory breaths are triggered by the patient. In contrast to CMV and ACV, however, in both IMV and SIMV the patient is allowed to breathe spontaneously between the mandatory breaths. In addition, with IMV and SIMV, the clinician can vary the ventilator support level according to the set IMV rate. At a high IMV rate, in which the patient's spontaneous effort is suppressed, IMV provides full ventilator support. At a zero IMV rate, it provides no support, and all breaths are spontaneous. Between these extremes, IMV provides partial ventilator support.

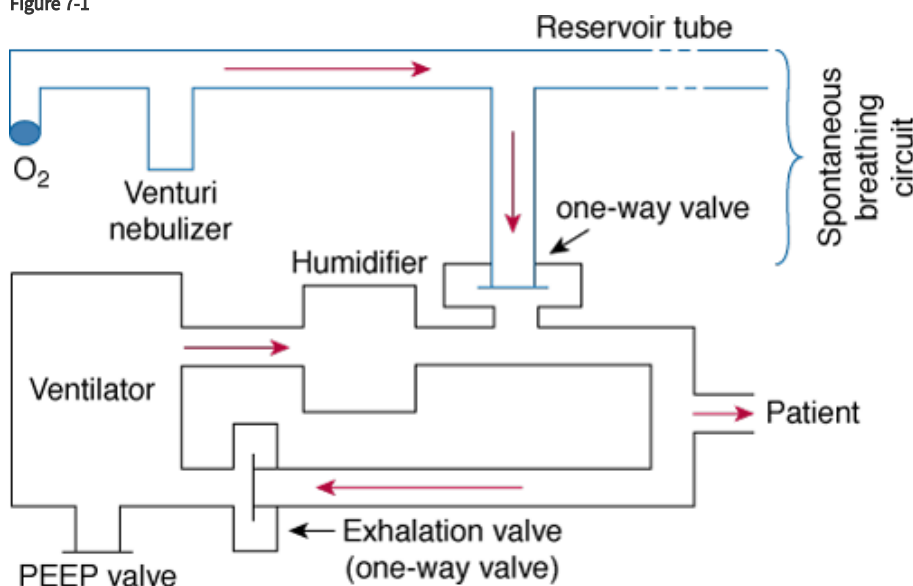
System Design

Three types of IMV systems are described: continuous-flow IMV and pressure-triggered and flow-triggered SIMV systems.

Continuous-Flow IMV

The original IMV design uses a continuous-flow system.³ Two parallel circuits—one for the patient's spontaneous breaths and the other for the mechanical breaths—are connected through a sidearm and a one-way valve, and share a common oxygen and air source. The continuous-flow IMV setup can be either an open or a closed system.¹⁷ The open system employs a reservoir tube that has a capacity of at least 1.5 times the patient's tidal volume (V_T) and is open to the atmosphere (Fig. 7-1). For this reason, continuous positive airway pressure (CPAP) cannot be applied during the spontaneous breathing cycles. To reduce inspiratory resistance, the side port of the spontaneous breathing circuit is placed between the patient Y and the humidifier. The continuous flow of fresh gas is humidified using a venturi nebulizer. The reservoir tubing's considerable length and the inability to maintain a CPAP level make this open system cumbersome.

Figure 7-1

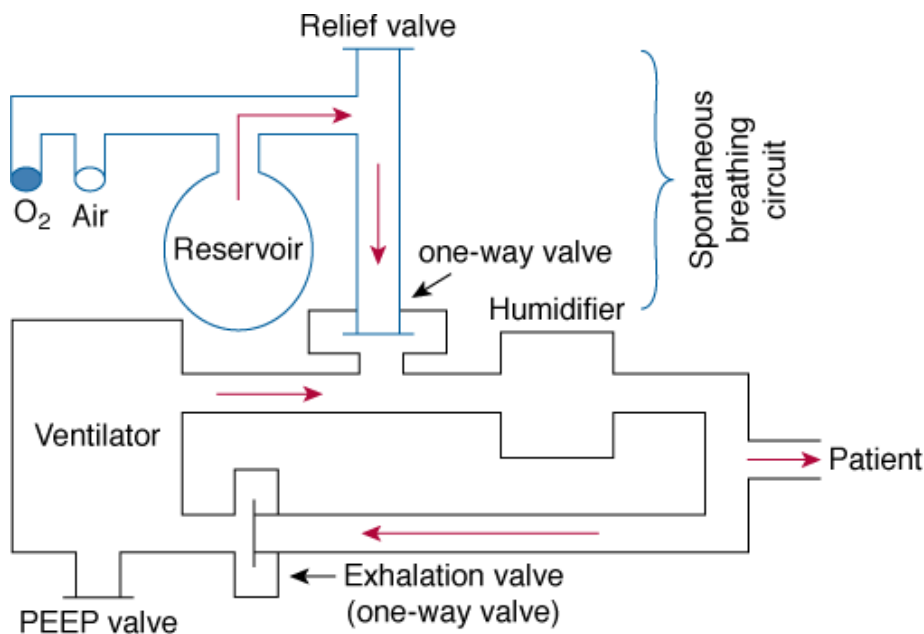


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Continuous-flow intermittent mandatory ventilation setup with a reservoir tube. The sidearm of the spontaneous breathing circuit is connected through a one-way valve to the inspiratory limb of the ventilator circuit. The sidearm is placed between the humidifier and the patient Y. See text for further explanation. *PEEP*, positive end-expiratory pressure.

The closed system employs a reservoir bag (Fig. 7-2) that minimizes airway pressure fluctuations because the inspired gas flow rate may be limited by the maximum flow generated by the hospital's compressed air and oxygen source. In addition, constant positive airway pressure can be maintained during both the mandatory (i.e., positive end-expiratory airway pressure [PEEP]) and spontaneous breaths (i.e., CPAP). During spontaneous breathing, the patient breathes from the reservoir bag via the one-way valve. When the ventilator cycles, the one-way valve closes, and a positive-pressure breath is delivered to the patient. During the mechanical breath, excess gas in the reservoir bag is vented through a relief valve. Exhalation occurs through the ventilator's exhalation circuit, which is supplied with a PEEP valve.

Figure 7-2



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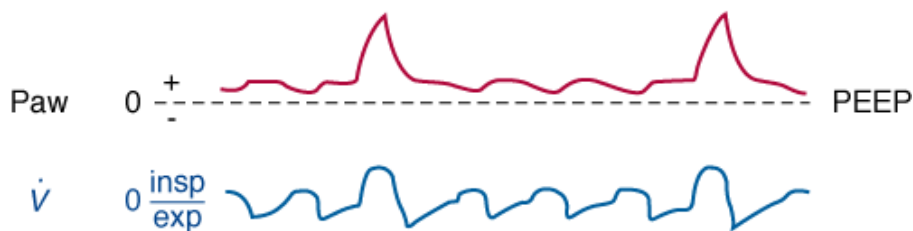
Continuous-flow intermittent mandatory ventilation setup with a reservoir bag. The sidearm for the spontaneous breathing circuit is connected through a one-way valve to the inspiratory limb of the ventilator circuit. The sidearm is placed proximal to the humidifier. See text for further explanation. *PEEP*, positive end-expiratory pressure.

In a continuous-flow IMV system, the inspired gas flow rate within the spontaneous breathing circuit must exceed the patient's peak inspiratory flow rate to minimize airway pressure fluctuations and hence the patient's inspiratory work. When set appropriately, spontaneous breathing in continuous-flow IMV should resemble breathing from the atmosphere.

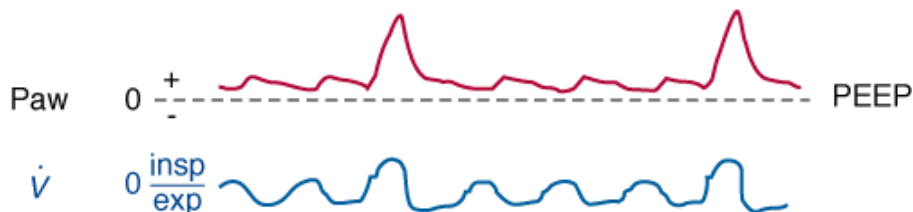
Continuous-flow IMV has several disadvantages.^{18–23} In addition to gas wastage, inaccurate V_T measurement and extra circuitry requirement, patient-ventilator asynchrony potentially can occur because mandatory breaths are not delivered in concert with the patient's inspiratory effort (Fig. 7-3A). Whereas asynchrony has no significant effect in adults,²⁴ in neonates, particularly preterm infants,²⁵ asynchrony associated with continuous-flow IMV resulted in large fluctuations of V_T ²⁶ and lower partial pressure of arterial oxygen (Pa_{O_2})^{27,28} than with SIMV. The effect of IMV on Pa_{O_2} was confirmed in a large randomized multicenter trial.²⁹ The degree and duration of hypoxia appear to result from active exhalation,³⁰ while muscle relaxant improves oxygenation.^{31,32} Furthermore, the application of SIMV is superior to IMV in terms of reduced duration of mechanical ventilation, requirement for reintubation, incidence of intraventricular hemorrhage, and bronchopulmonary dysplasia.³³ In term infants and children (ages 1 month to 4 years), however, IMV or SIMV plus pressure support has similar effects on duration of mechanical ventilation, weaning, and intensive care unit length of stay.³⁴

Figure 7-3

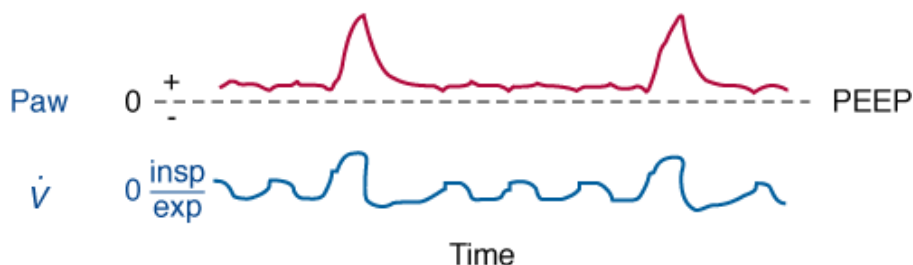
A. Continuous-flow IMV



B. Pressure-triggered SIMV



C. Flow-triggered SIMV



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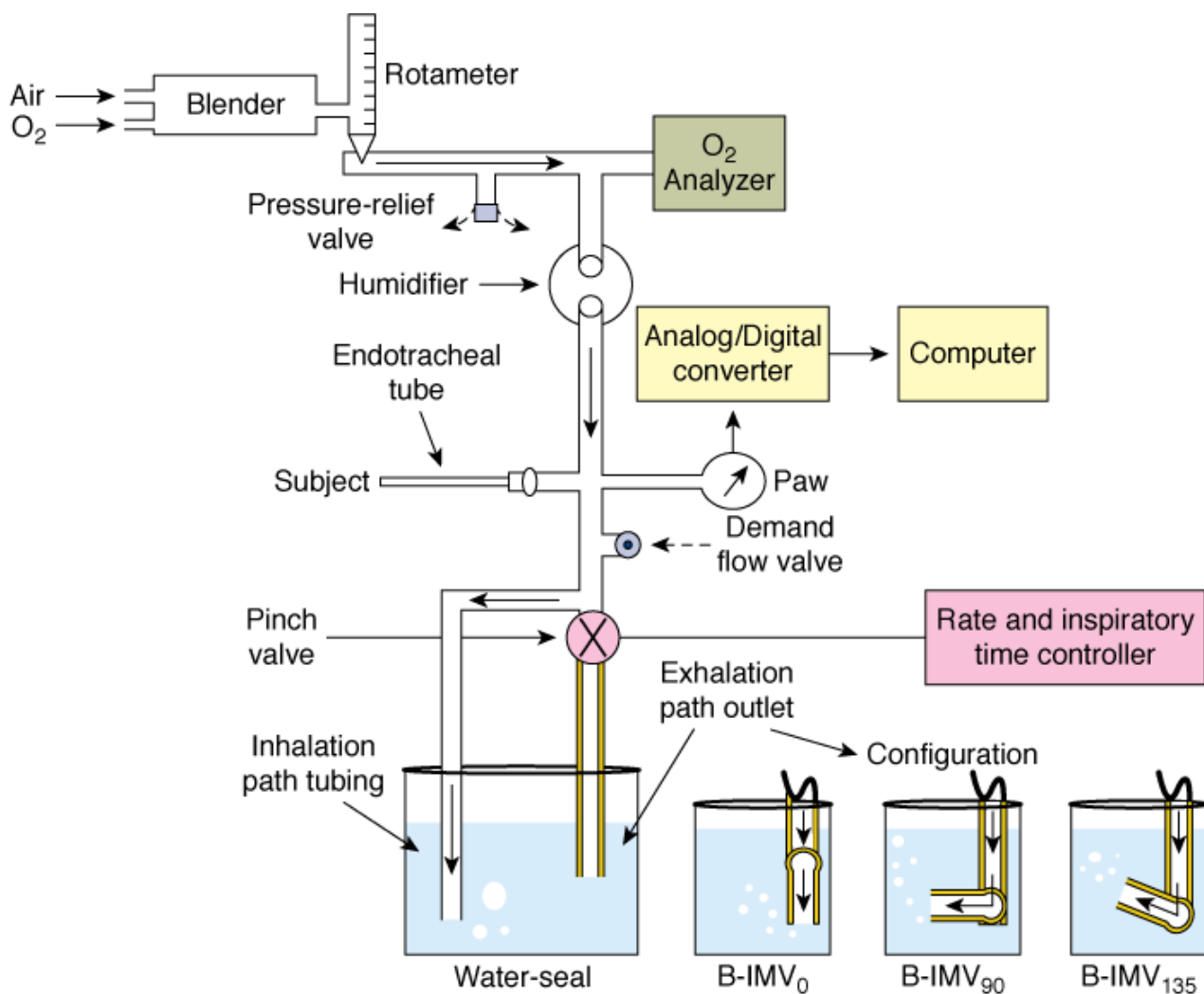
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A. Continuous-flow intermittent mandatory ventilation (IMV). Airway pressure (P_{aw}) and flow tracings ($\dot{V}$). The vertical *dashed line* indicates the onset of inspiratory flow. Minimal fluctuations of P_{aw} are a result of circuit resistance. Large fluctuations of P_{aw} occur when inspiratory flow rates are insufficient and a reservoir bag is not used. *exp*, Expiration; *insp*, inspiration; *PEEP*, positive end-expiratory airway pressure. **B.** Pressure-triggered synchronous intermittent mandatory ventilation (SIMV). A period of zero flow before the onset of flow (indicated by the vertical *dashed line*) can be detected on the flow tracing. Zero flow coincides with patient triggering to open the proportional valve. During the unassisted breathing after flow onset, P_{aw} continues to drop during inspiration because of inadequate flow delivery. **C.** Flow-triggered synchronous intermittent mandatory ventilation (SIMV). Triggering phase is significantly shorter than with pressure-triggered SIMV. During the unassisted breathing following patient triggering, P_{aw} is maintained at or slightly above the PEEP level, suggesting adequate flow delivery during inspiration.

A technologically advanced continuous-flow IMV or CPAP is the flow-regulated IMV or CPAP.³⁵ Flow within the circuit is polled, for example, every 20 minutes and used as a feedback signal to increase the basal flow to match the patient's ventilatory demand during inspiration and subtract it during exhalation. Flow-regulated IMV or CPAP eliminates airway fluctuations and decreases imposed work during both inspiration and exhalation. Using a mechanical lung model with flow-regulated CPAP of 5 cm H₂O, the total imposed work was 3.4 mJ/breath versus 43.5 mJ/breath with a continuous-flow CPAP device and an added 20-L reservoir bag.³⁵

DiBlasi et al³⁶ described a modified continuous-flow IMV, a combination of continuous-flow IMV and low-to-moderate frequency oscillatory ventilation. It is also called Bubble IMV (B-IMV), intended as an inexpensive means of ventilator support in infants. The device is equipped with a microprocessor-controlled rate, inspiratory timing, and pinch valve. The pinch valve regulates the path of gas exiting the system into a water-seal chamber (Fig. 7-4). The inhalation path controls the peak inspiratory pressure level of the mandatory breaths, and the exhalation path allows spontaneous breaths (i.e., CPAP), and control PEEP level. The exhalation path outlet is connected to an adjustable bubbler with three configurations: a bubbler angle of 0 degrees, 90 degrees, and 135 degrees. The bubbler angle determines the oscillation amplitudes in airway pressure (P_{aw}). Only the bubbler angle of 135 degrees is associated with the largest pressure oscillations at a frequency of 2 to 5 Hz, and, hence, the largest change in volume (Fig. 7-5A).³⁷ In paralyzed juvenile rabbits with saline lavage-induced lung injury, B-IMV provided comparable peak inspiratory and mean P_{aw} , PEEP, and Pa_{O_2} as with CMV, irrespective of the bubbler angle (Fig. 7-5B).³⁶ B-IMV₁₃₅, however, was associated with consistently low mean partial pressure of arterial carbon dioxide (Pa_{CO_2} ; 35 mm Hg). Mean Pa_{CO_2} with B-IMV₉₀, B-IMV₀, and CMV was 45, 45, and 55 mm Hg, respectively. The low Pa_{CO_2} with B-IMV₁₃₅ is caused by additional volume produced by the oscillatory pressures (DiBlasi, personal communication) (Fig. 7-5A). At this time, B-IMV is an experimental ventilator, its potential application in preterm infants with respiratory distress syndrome and its efficacy when compared with SIMV remains to be determined.

Figure 7-4

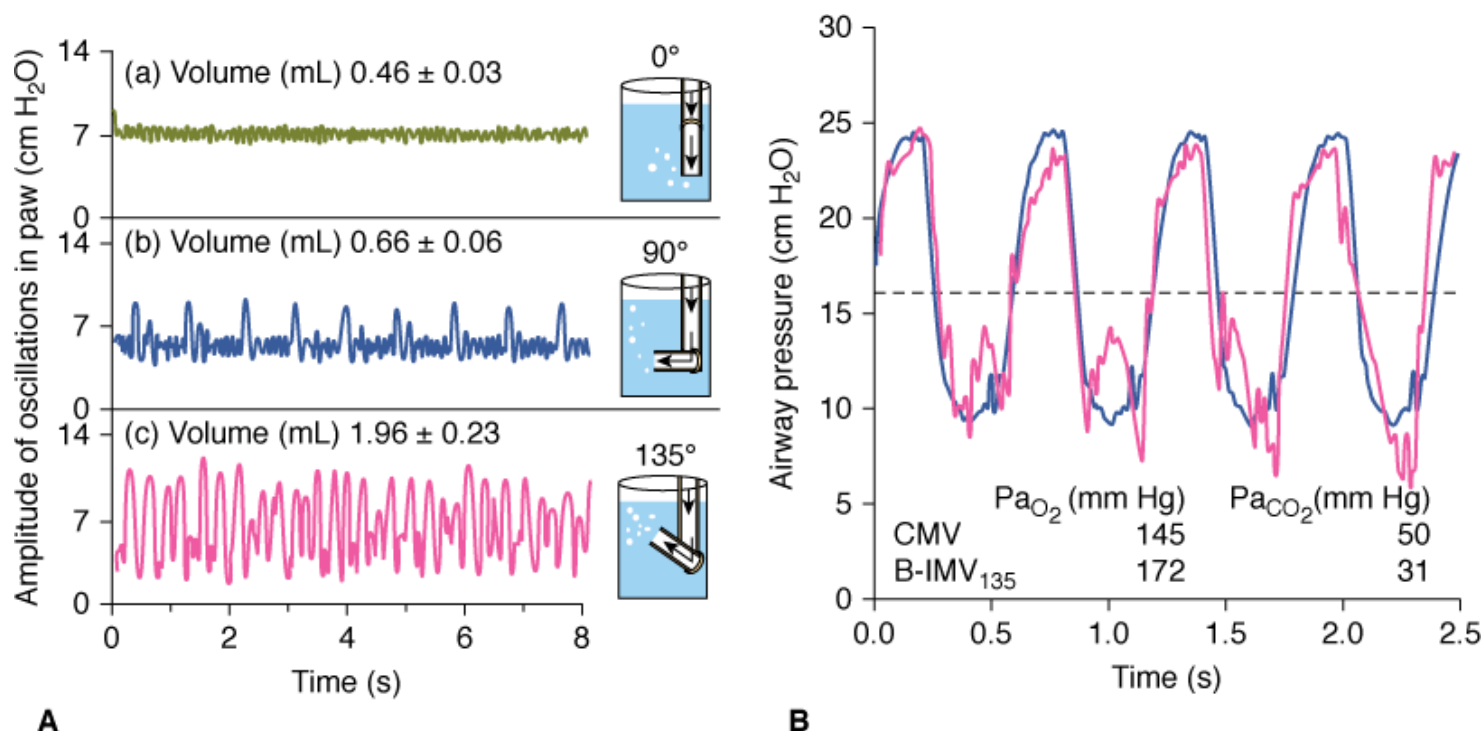


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Bubble-intermittent mandatory ventilation (B-IMV). A modified continuous-flow IMV is equipped with a rotameter to control bias flows of air and oxygen. In line of patient's circuit are a high-pressure relief valve set to a limit of 40 cm H₂O, oxygen analyzer, humidifier, airway pressure monitor, a demand flow valve that allows room air to enter the circuit if the bias flow does not meet the patient's flow demand, and a pinch valve with a rate and inspiratory time controller. The pinch valve controls the path of gas exiting the system, either through the inhalation path with the tube placed deep in the water-seal chamber to control peak inspiratory pressure level of the mandatory breaths, or through the exhalation path to allow spontaneous breaths (continuous positive airway pressure [CPAP]) and control positive end-expiratory airway pressure (PEEP) level. The exhalation path outlet is connected to a bubbler with the angles adjustable to 0 degrees, 90 degrees, and 135 degrees (B-IMV₀, B-IMV₉₀ and B-IMV₁₃₅, respectively). (Adapted, with permission, from DiBlasi RM, Zignego JC, Smith CV, et al. Effective gas exchange in paralyzed juvenile rabbits using simple, inexpensive respiratory support devices. *Pediatr Res*. 2010;68:526–530.)

Figure 7-5



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A. Effects of bubbler angle configuration on airway pressure (P_{aw}) oscillations. Bubble-continuous positive airway pressure connected to a test lung with a bias flow set at 8 L/min. Bubbler angle of 135 degrees produces the largest airway pressure oscillations at a frequency of 2 to 5 Hz, and largest change in volume. **B.** Superimposed airway pressure waveform during CMV (solid black) and bubble-intermittent mandatory ventilation with bubble angle of 135 degrees (B-IMV₁₃₅) (double gray line) in an animal with lung injury. Horizontal dashed line represents mean airway pressure. (Adapted, with permission, from DiBlasi RM, Zignego JC, Tang DM, et al. Noninvasive respiratory support of juvenile rabbits by high-amplitude bubble continuous positive airway pressure. *Pediatr Res.* 2010;67:624–629.)

Pressure-Triggered and Flow-Triggered SIMV

SIMV is currently the standard for clinical use.^{10,11} SIMV incorporates a demand valve that is triggered by the patient with each spontaneous breath and delivers a mandatory breath in concert with the patient's inspiratory effort. If the patient ceases to trigger the ventilator, mandatory breaths will be triggered by the machine and delivered according to the preset rate. The demand valve can be triggered by either a fall in pressure (pressure-triggered) or a change in flow (flow-triggered). In pressure-triggered SIMV, a preset pressure sensitivity must be achieved before the ventilator delivers fresh gas into the inspiratory circuit.^{38,39} A noticeable delay in opening the demand valve occurs between onset of inspiratory effort and flow delivery (see Fig. 7-3B). Flow-triggered SIMV uses a preset flow sensitivity as the trigger mechanism.^{38,39}

The pressure-triggering and flow-triggering characteristics of the spontaneous breaths (CPAP) are an important component of the imposed work of a SIMV system (see Chapter 3).^{40,41} This situation arises because there is little adaptation to the mandatory breaths' ventilatory assistance unless the system is set at a substantial assistance level.⁴² Fortunately, most modern microprocessor-based ventilators employ a remarkably responsive proportional solenoid valve such that the work imposed during the trigger phase (the interval from onset of patient effort to valve opening or flow delivery) is a small percentage of the total inspiratory work of breathing (<10% with pressure-triggered SIMV).⁴³

Nevertheless, flow triggering has become the preferred triggering method not only for SIMV but also for various other ventilation modes because of its faster response time and therefore shorter triggering phase compared with pressure triggering (see Fig. 7-3C).⁴⁴ In the posttrigger phase (the interval from onset of flow delivery to the end of inspiration), flow delivery with pressure triggering may not be adequate during the spontaneous breaths.³⁸ The feedback signal for flow delivery is the gradient between a manufacturer-determined target pressure and circuit pressure. As the pressure gradient increases, flow delivery increases. This pressure gradient can be enhanced by adding pressure support or sensing the circuit pressure at the distal end of the endotracheal tube.^{45,46} From a practical standpoint, adding pressure support is preferable. Alternatively, flow delivery during the spontaneous breaths may be augmented by using changes in flow instead of pressure as a feedback signal for adding and subtracting flow from the base flow during inspiration and exhalation, respectively.³⁵

The mandatory breaths can be in the form of volume-limited or pressure-limited. In preterm infants, pressure-targeted ventilation is prevalent,¹⁴ with similar efficacy as with volume-targeted ventilation in maintaining oxygenation.⁴⁷ Augmenting flow delivery during the mandatory breaths can be accomplished by setting the pressure-attack rate sufficiently high when pressure-limited mandatory breaths are employed.^{48,49} To maintain a constant V_T with pressure-limited mandatory breaths, both pressure-limited and volume-limited breaths can be combined in the form of dual control within breaths⁵⁰ or breath-to-breath ventilation (see Chapter 15).⁵¹ Several microprocessor-based ventilators are equipped with dual-control breath-to-breath ventilation that can be applied in the SIMV mode.⁵² In essence, this form of mandatory breath is a pressure-limited, time-cycled breath that uses V_T as feedback control for continuously adjusting the pressure limit to attain the set V_T . The volume signal used as feedback to the ventilator controller is the volume exiting the ventilator and not the exhaled V_T . This step prevents runaway of airway pressure that could occur if a leak in the circuit caused inaccurate measurement of exhaled volume. As an additional safety feature, if the volume exiting the ventilator exceeds 150% of the set V_T , then the ventilator exhalation valve opens, ending the mechanical inspiration. With SIMV, most microprocessor-based ventilators are able to apply pressure support and dual-control breathing to the spontaneous and mandatory breaths, respectively, to improve flow delivery and maintain a set V_T during pressure-limited mandatory breaths. In preterm infants, because changes in respiratory system mechanics occur frequently, the set V_T of the mandatory dual-control breaths not only provides a guaranteed volume when respiratory system compliance decreases but also prevents overinflation when compliance improves.^{53,54} In fact, a recent report demonstrates a decrease in mortality and chronic lung disease with volume-targeted compared with pressure-targeted ventilation.⁵⁵

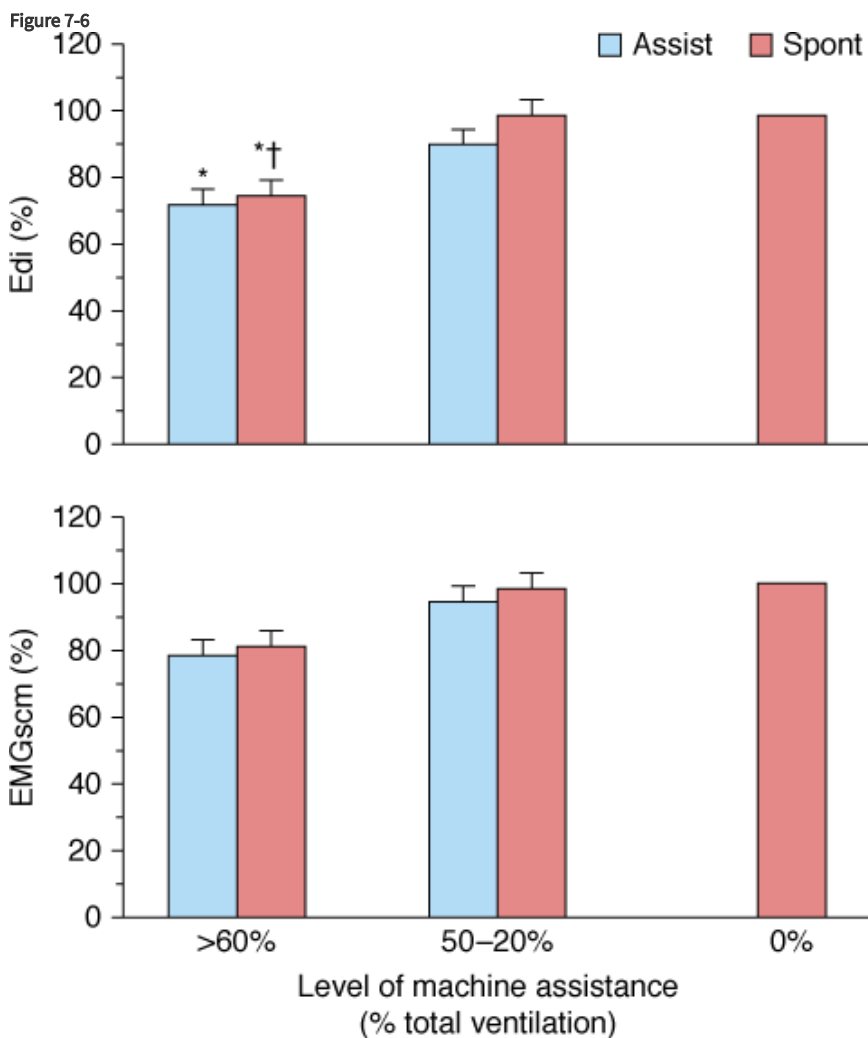
Physiologic Effects

The physiologic effects of IMV not listed in this section are discussed in comparison with other ventilation modes.

Control of Breathing and Breathing Patterns

In early IMV development, it was thought that inspiratory muscle activity is downregulated during the mandatory breaths and that IMV allows a combination of unassisted breathing with respiratory muscle rest to promote weaning.^{5,56} By adjusting mandatory breath frequency, inspiratory effort could be modified until the patient resumed complete control of spontaneous breathing. Several studies, however, disproved this hypothesis.^{42,57–60} Imsand et al⁵⁷ studied the

neuromuscular output of patients recovering from acute exacerbation of chronic obstructive pulmonary disease who were receiving pressure-triggered SIMV. The mandatory-breath V_T was set at 10 to 12 mL/kg. Neuromuscular output was estimated from the amplitude of the integrated diaphragmatic electrical activity (*Edi*), measured using bipolar esophageal electrodes. Sternocleidomastoid muscle electrical activity was recorded using surface electrodes. Neuromuscular output, occlusion esophageal pressure ($P_{es0.1}$, another index of neuromuscular output), and neural inspiratory time (T_I) were measured at three ventilator support levels: >60%, 50% to 20%, and 0% of total support. Total support was defined as the support at which *Edi* was suppressed completely. Only at the highest machine assistance rate did *Edi* decrease significantly, whereas sternocleidomastoid muscle electrical activity did not (Fig. 7-6). Moreover, across all levels of ventilator support, both *Edi* (Fig. 7-7) and $P_{es0.1}$ of the unassisted and machine breaths were similar. $P_{es0.1}$, however, tended to increase with decreasing machine assistance. Likewise, neural T_I , defined as the interval from onset to peak *Edi*, was equivalent for unassisted and machine breaths.



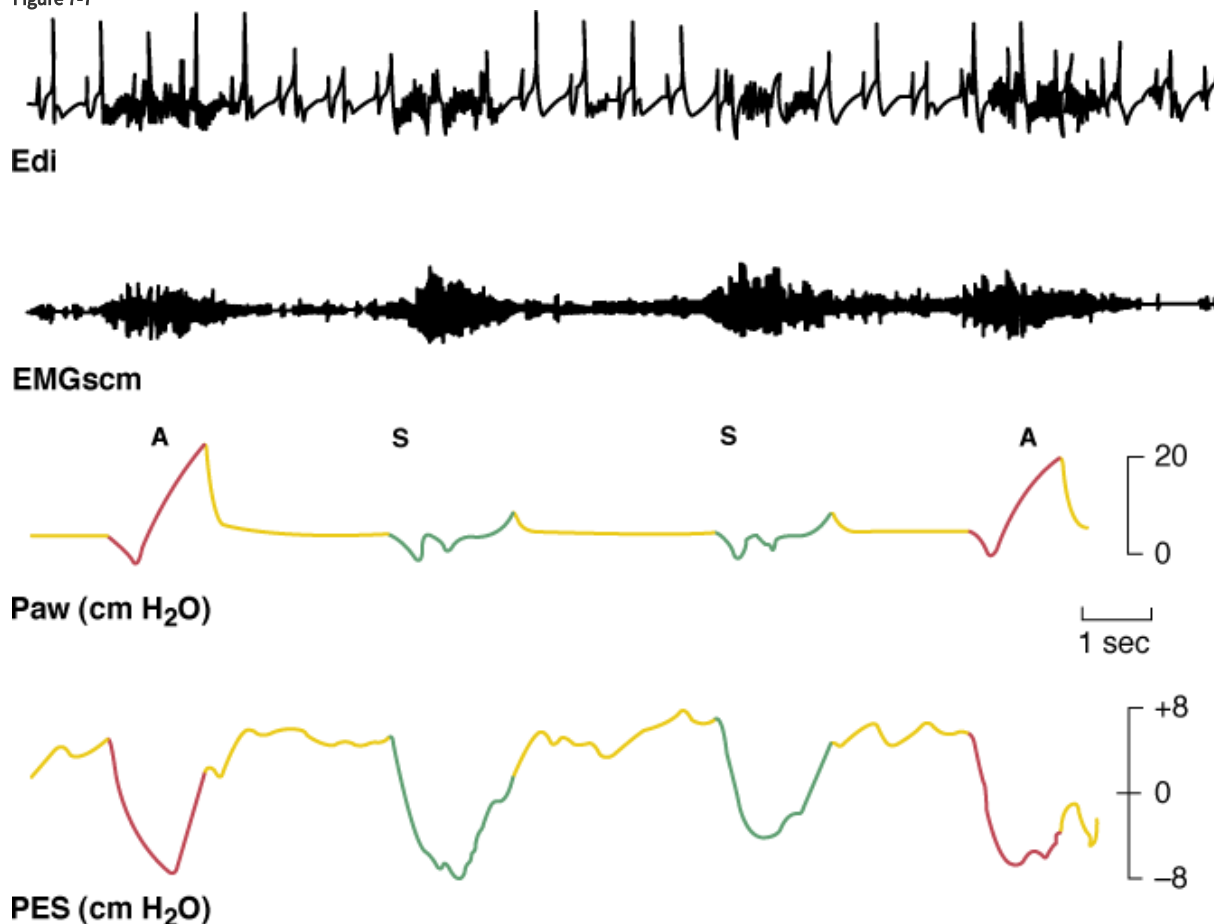
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Peak inspiratory amplitude of integrated electrical activity of the diaphragm (*Edi*) and sternocleidomastoid muscles (*EMGscm*) at three levels of machine assistance during SIMV, expressed as a percentage of mean value of 0% of or minimal (4 breaths/min) machine assistance. Values are mean $\pm$ standard error. * $P < 0.01$ compared with 0% of machine assistance. † $P < 0.05$ compared with spontaneous cycles at 50% to 20% of assistance. *Assist*, Machine-assisted breaths; *spont*, intervening spontaneous breaths. (Adapted, with permission, from Imsand C, Feihl F, Perret C, et al. Regulation of

inspiratory neuromuscular output during synchronized intermittent mechanical ventilation. *Anesthesiology*. 1994;80:13–22.)

Figure 7-7



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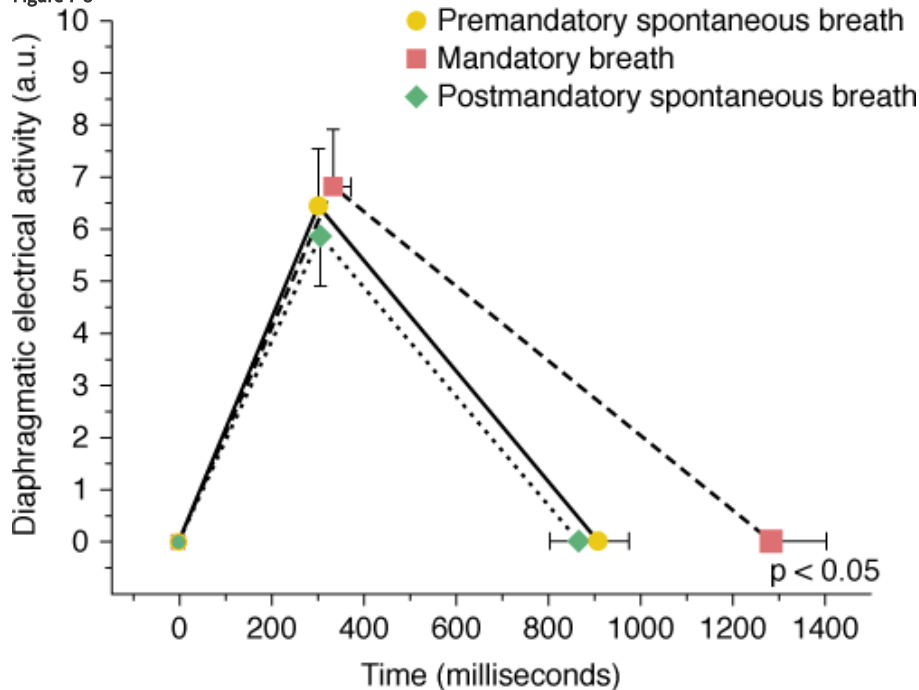
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Electrical activity of the diaphragm (*Edi*) and sternocleidomastoid muscles (*EMGscm*) in a patient receiving SIMV. Intensity and duration of electrical activity in successive assisted (*A*) and intervening spontaneous (*S*) breaths are similar. The regular spikes in the *Edi* tracing are the QRS complex of the electrocardiogram signal. *Paw*, Airway pressure; *Pes*, esophageal pressure; *red lines* and *green lines*, inspiratory phase of the assisted and spontaneous breaths, respectively; *yellow lines*, expiratory phase. (Adapted, with permission, from Imsand C, Feihl F, Perret C, et al. Regulation of inspiratory neuromuscular output during synchronized intermittent mechanical ventilation. *Anesthesiology*. 1994;80:13–22.)

In another study, Uchiyama et al⁵⁸ applied continuous-flow IMV to anesthetized rabbits and measured *Edi* with implanted electrodes into the diaphragm. The IMV rate was set at 0, 5, 10, 15, and 20 breaths/min, at which *Edi* was suppressed completely. *Edi* was expressed as percent of *Edi* at zero IMV rate per minute. Compared with spontaneous breathing, *Edi* decreased significantly only at an IMV rate of 15 breaths/min (75% of total support). In contrast with the study of Imsand et al,⁵⁷ at an IMV rate of 10 and 15 breaths/min, *Edi* of the mandatory breaths was absent. The difference in neuromuscular output response during mandatory breaths in the study of Uchiyama et al⁵⁸ may be related to the effect of anesthesia and the large *V_T* applied (15 mL/kg), causing vagally mediated inspiratory inhibition.⁵⁹ It appears that in anesthetized animals at ventilator support of 50% or greater, breath-by-breath adaptation to ventilator assistance occurred, but it was not observed in conscious humans.

In unsedated preterm infants with acute respiratory failure receiving flow-triggered SIMV, Beck et al⁶⁰ measured Edi and neural timing of mandatory breaths and the unassisted breaths immediately preceding and following the mandatory breaths using miniaturized electrodes mounted on a feeding tube. The set SIMV rate ranged from 5 to 25 breaths/min. Both Edi and neural T_I amplitude were similar for the mandatory and unassisted breaths (Fig. 7-8). In contrast to improved triggering in adults,⁵⁷ during the mandatory breaths, a substantial delay was observed from onset of Edi to flow delivery, an average of 95 milliseconds (range: 5 to 110 milliseconds). Because most of the infants had bronchiolitis and likely were hyperinflated, the authors attributed the trigger delay or inspiratory asynchrony to neuroventilatory uncoupling (a delay from onset of diaphragmatic activation to flow generation)⁶¹ rather than to the ventilator triggering system. Neural expiratory time (T_E) of the mandatory breath, defined as the interval from peak to Edi onset of the subsequent breath, was prolonged. The relative increase in neural T_E during the mandatory breaths was related to the time the ventilator continued to inflate beyond the end of neural T_I ($R^2 = 0.66$; $P = 0.01$). This extended time was consistent with a vagally mediated reflex when pulse inflation was delivered into early expiration.^{62,63} Both inspiratory asynchrony and expiratory asynchrony were present during every mandatory breath and constituted, on average, 53% of the total breath duration (Fig. 7-9). Expiratory asynchrony may result in increased peak airway pressure as the subject attempts to terminate inspiration by recruiting the expiratory muscles.⁶⁴ It also may result in increased work of breathing and discomfort and manifest as a patient “fighting the ventilator,”⁶⁵ potentially dictating sedation.

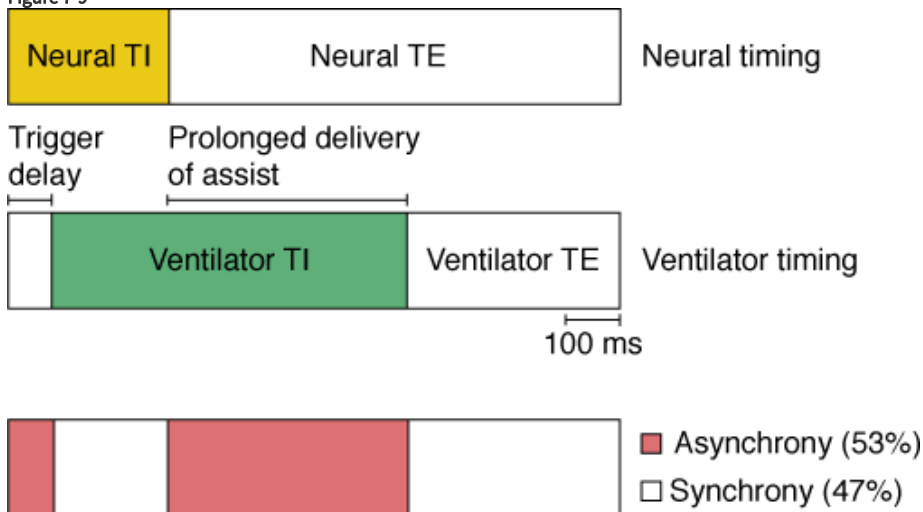
Figure 7-8



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Diaphragmatic electrical activity (Edi) time profile for group mean data. No significant difference exists between peak Edi amplitude and neural timing on inspiration for the premandatory spontaneous breaths (*solid line*), mandatory breaths (*dashed line*), or after mandatory spontaneous breaths (*dotted line*). Neural expiratory time was prolonged significantly for the mandatory breaths (*dashed line*). The plot does not represent the shape of the Edi recruitment pattern but simply represents three points: the onset of Edi, peak of Edi, and onset of Edi of subsequent breath. Values are mean $\pm$ standard error. (Adapted, with permission, from Beck J, Tucci M, Emeriaud G, et al. Prolonged neural expiratory time induced by mechanical ventilation in infants. *Pediatr Res*. 2004;55:747–754.)

Figure 7-9



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Inspiratory and expiratory asynchrony during mandatory breaths. Schematic of patient neural timing (*upper bar*) and ventilator timing (*middle bar*) during mandatory breaths. Upper bar shows the neural inspiratory time (T_I) (yellow area) and neural expiratory time (T_E) (white area). Middle bar shows periods of ventilator timing, including trigger delay, ventilator T_I (green area), and ventilator T_E . Bottom bar shows periods of infant-ventilator synchrony (white area) and asynchrony (red area). (Adapted, with permission, from Beck J, Tucci M, Emeriaud G, et al. Prolonged neural expiratory time induced by mechanical ventilation in infants. *Pediatr Res*. 2004;55:747-754.)

Using airway occlusion pressure ($P_{0.1}$) and mean inspiratory flow rate (V_T/T_I) as indices of neuromuscular output, Weiss et al⁶⁶ examined the effect of IMV on respiratory drive in stable ventilator-dependent patients who received mechanical ventilation for 12 to 24 hours each day. The IMV rate was initially set at 10 breaths/min and reduced gradually to 6, 3, 2, and 1 breath/min, and each level was maintained for 10 minutes. The V_T of the mandatory breaths was set at 8 to 12 mL/kg. End-tidal partial pressure of carbon dioxide (P_{CO_2}) and arterial oxygen saturation did not change as the IMV rate was reduced. In the presence of stable chemical input, $P_{0.1}$ and V_T/T_I measured during spontaneous breathing varied inversely with IMV rate. Increases in V_T/T_I with IMV rate reduction primarily were the result of progressive V_T augmentation because inspiratory time (T_I) and the duty cycle (T_I/T_{TOT}) remained unchanged at all IMV support levels. This V_T response probably was related to changes in CO_2 production that stimulated chemoreceptors. In acutely ill patients,⁴² the effect of IMV on $P_{0.1}$, measured during spontaneous breathing, is similar to that in stable patients.⁶⁶ Using another index of neuromuscular output, dP/dt (change in pressure over change in time) of esophageal pressure at the onset of ventilator triggering, Leung et al⁴³ also demonstrated an inverse relationship between dP/dt and machine-assistance level.

As with neuromuscular output, IMV's effects on breathing patterns are determined by assistance level. Both V_T and unassisted breath frequency increase progressively with a decreasing IMV rate.^{42,43,57,67} Because V_T of the unassisted breaths is relatively small compared with the mandatory breaths, dead-space-to-tidal-volume ratio inevitably increases. Consequently, spontaneous breathing frequency increases to maintain constant alveolar ventilation. Indeed, Pa_{CO_2} remained constant at all machine-assistance levels.⁶⁷

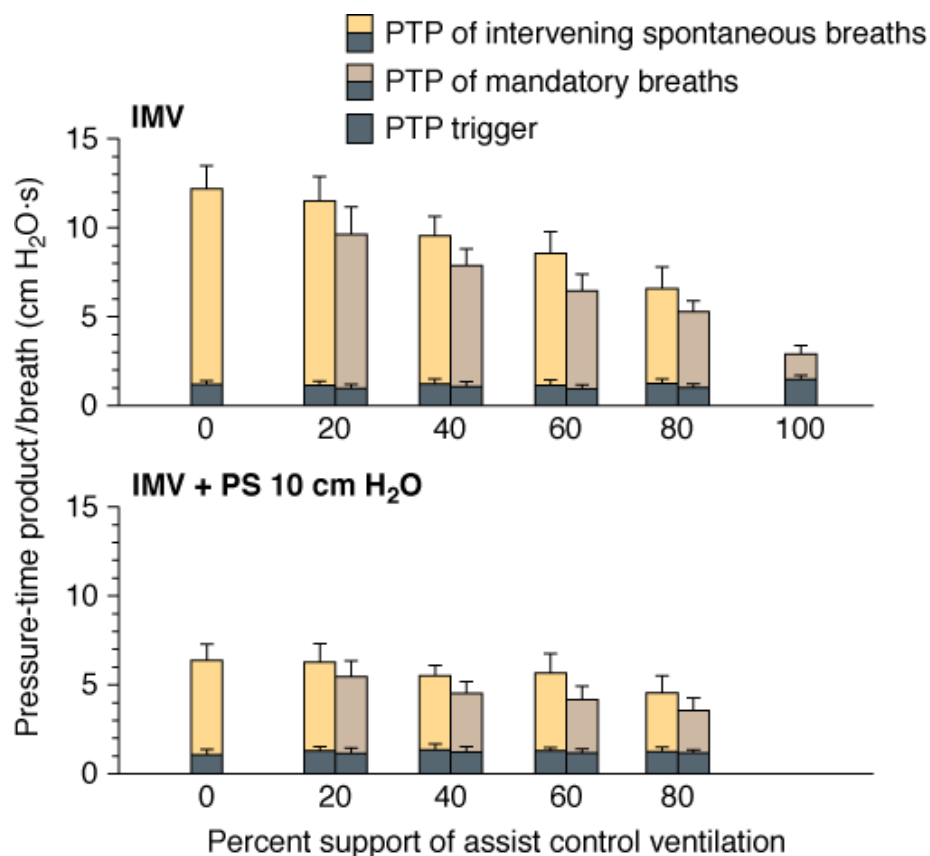
The sensation of dyspnea during SIMV depends on the assistance level when applied without pressure-support ventilation (PSV) but was independent of assistance level when 10 cm H₂O of pressure support was added to the unassisted breaths.⁴³ During low assistance levels, dyspnea probably is secondary to increased inspiratory effort, whereas the high rate of nontriggering efforts may be responsible for the dyspnea at higher assistance levels. Knebel et al⁶⁸ also demonstrated that during the weaning trial on SIMV, the sensation of dyspnea and patient anxiety were independent of support level. In that study,⁶⁸ a PEEP of 5 cm H₂O was maintained, which might have offset the increased inspiratory muscle work with a decreasing IMV rate.^{69,70}

In summary, with IMV, the neuromuscular output of both the mandatory and unassisted breaths is downregulated only at high machine-assistance levels (approximately 75% to 80% of total machine assistance).^{57,58} Breath-by-breath adaptation to machine assistance does not occur. The fact that Edi was the same during assisted and preceding unassisted breathing suggests that neuromuscular output during the mandatory breaths must be determined at least in part by factors operating during the previous breath. This could be accounted for by the central neural feedback mechanism in the manner of a “memory” phenomenon as described by Eldridge⁷¹ (see below). Neuromuscular output increases inversely with the set IMV rate. Whereas V_T and frequency increase when IMV rate decreases, neural inspiratory timing of the mandatory and unassisted breaths is of the same duration across all levels of machine assistance. Neural expiratory timing of the mandatory breaths is prolonged secondary to mechanical T_I encroaching into neural T_E , causing expiratory asynchrony. Respiratory timing asynchrony, however, can be substantially mitigated by using inspiratory and expiratory neural triggering with a neurally adjusted ventilatory assist.^{61,72,73}

Work of Breathing and Inspiratory Effort

The effect of IMV on inspiratory muscle work and inspiratory effort is, as expected, determined by the level of machine assistance, and the response is similar in adults^{42,43,57} and neonates.⁶⁷ A higher number of mandatory breaths reduces total workload and inspiratory effort in patients during acute respiratory failure⁴² or recovery.^{43,74,75} Marini et al⁴² measured work of breathing per liter of ventilation (joules per liter) and pressure-time product per breath during ACV and at SIMV support levels of 80%, 60%, 40%, 20%, and 0% of the ACV rate. When the SIMV support level was less than 80% of the ACV rate, the patient’s total workload increased markedly compared with that during ACV. Moreover, at all SIMV levels, the patient expended a similar effort during both the mandatory and spontaneous breaths, as gauged by pressure-time product per breath, which reflects the energy consumed by contracting respiratory muscles (patient effort). The patient does not vary effort on a breath-by-breath basis in response to machine unloading, as confirmed by others on the basis of Edi recording.^{57,60} It appears that during SIMV the heightened respiratory activity during the spontaneous breath is carried over to the subsequent mandatory breath in the manner of an “after discharge” or “memory” phenomenon.⁷¹ In fact, when 10 cm H₂O of pressure support was applied during the intervening spontaneous breaths, inspiratory muscle effort decreased not only during the intervening breaths but also during the mandatory breaths (Fig. 7-10).⁴³ This afterdischarge phenomenon is central in origin and unrelated to chemoreceptor or mechanoreceptor feedback or the state of sleep or wakefulness.^{71,76}

Figure 7-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Pressure-time product (PTP) per breath during synchronous intermittent mandatory ventilation (IMV) in the presence and absence of pressure support (PS). The addition of 10 cm H₂O PS to the intervening spontaneous breaths decreased the overall PTP per breath. PTP per breath, however, did not differ between the mandatory and intervening spontaneous IMV breaths at each level of machine assistance, irrespective of whether PS was present or absent. (Adapted, with permission, from Leung, et al. *Am J Respir Crit Care Med*. 1997;155:1940–1948. Reprinted with permission of the American Thoracic Society. Copyright © American Thoracic Society.)

The patient workload during IMV is determined not only by machine-assistance level, but also by the IMV system employed.^{74,75} Patient workload or inspiratory effort during spontaneous breathing cycles with flow-triggered SIMV was significantly less than with a pressure-triggered SIMV system,^{74,75} particularly at a relatively low SIMV rate.⁷⁵ The effect, however, of the triggering system on patient mandatory breath workload is conflicting. Sassoon et al⁷⁵ demonstrated that the triggering system had no effect on patient mandatory-breath workload. In contrast, Giuliani et al⁷⁴ found that patient mandatory-breath workload was significantly less with flow-triggered than with pressure-triggered SIMV irrespective of the mandatory-breath type (whether SIMV was flow-limited or pressure-limited). Moreover, patient workload with combined flow-triggered and pressure-limited mandatory breaths was lower than the other combinations of triggering and mandatory-breath types (flow-triggered flow limit and pressure-triggered flow and pressure limit).⁷⁴ The decrease was so substantial that the pressure-time product per breath of the mandatory breath was significantly less than that of the intervening spontaneous breaths. It is not clear as to why the data of Giuliani et al⁷⁴ differed from the data of others^{42,43,57,60} who demonstrated the lack of breath-by-breath adaptation to machine unloading.

In summary, during acute respiratory failure or recovery, patient workload during SIMV remains substantial unless the SIMV support is equivalent to or greater than 80% of the support during ACV.^{42,43} IMV often allows adequate gas exchange at lower support levels⁷⁷ (see the section [Intermittent Mandatory Ventilation and Assist-Control Ventilation](#)) but at the expense of increased patient workload.^{42,43,57}

Rationale, Advantages, and Limitations

IMV was developed as an alternative to [CMV](#) and ACV. The rationale for developing IMV was based on the premise that by maintaining spontaneous breathing amid the mechanical breaths, (a) machine breaths can be titrated to adjust alveolar ventilation, thereby decreasing the incidence of respiratory alkalemia,⁷⁸ (b) distribution of inspired gas and ventilation-perfusion to the lung base improves, and physiologic dead space decreases,⁷⁹ (c) mean airway pressure and pulmonary vascular resistance decrease, allowing PEEP titration and therefore more effective treatment of hypoxemia,⁸⁰ and (d) pleural pressure decreases, resulting in better venous return and cardiac output.⁷⁹ Unfortunately, subsequent clinical trials did not support some of the rationales (see the section [Intermittent Mandatory Ventilation and Controlled Mechanical Ventilation](#)).

IMV's flexibility in providing a range of support levels makes it advantageous for use as a primary means of ventilator support⁸¹ and as a method for discontinuing mechanical ventilation.⁵ An inherent limitation of IMV is the fixed rate of machine breaths. When set at a very high rate, IMV functions as [CMV](#) that has been shown to rapidly lead to diaphragmatic muscle atrophy in both experimental animals^{82,83} and humans.^{84,85} During weaning from mechanical ventilation, the pace of decreasing the IMV rate potentially can prolong weaning when it is inappropriately slow.

Indications and Contraindications

IMV has been used as a primary means of ventilator support in postoperative patients⁸⁶ and during acute respiratory failure of various etiologies.^{6,7,87-91} For treatment of acute respiratory failure in adults, North Americans apply SIMV with pressure support more frequently than ACV (49% vs. 27%, respectively).¹² A similar trend is observed in Australia and New Zealand, 41% versus 17% with other mandatory pressure-limited modes of ventilation.¹³ In preterm neonates, the frequency of SIMV application in acute respiratory failure is higher than in adults, 60% versus 31% with ACV.¹⁴ However, it needs to be emphasized that in preterm infants in addition to the ventilatory strategy used, early treatment with surfactant is essential.⁹² The level of machine assistance must be tailored to the patient's ventilatory requirement. The optimal mandatory-breath settings are similar in principle to those for ACV.⁹³⁻⁹⁵ A low IMV setting is contraindicated when the patient's ventilatory demand is high; likewise, the setting should not be left high after the patient's ventilatory requirement has decreased. A gradual decrease in the IMV rate is unwarranted when discontinuation from mechanical ventilation is imminent.

Comparison with Other Modes

Intermittent Mandatory Ventilation as a Primary Means of Ventilator Support

Intermittent Mandatory Ventilation and Controlled Mechanical Ventilation

Unlike IMV, CMV imposes a fixed breathing pattern. Hence, to some extent, a comparison favors IMV, which has been claimed to be superior to CMV for the following reasons: (a) it prevents the patient from “fighting the ventilator,” reducing the need for sedation and paralysis; (b) it prevents respiratory alkalosis; (c) it improves intrapulmonary gas distribution; (d) it lowers mean airway pressure, benefiting cardiac output and preventing barotrauma; (e) it decreases oxygen consumption ($\dot{V}_{O_2}$); (f) it prevents muscle atrophy and discoordination; and (g) it improves renal function.

1. *IMV prevents the patient from “fighting the ventilator,” reducing the need for sedation and paralysis.* Because the patient has no autonomy to alter breathing pattern with CMV, the patient will increase inspiratory effort, which is manifested as “fighting the ventilator” when ventilatory demand rises.⁷⁸ No studies, however, have compared the respective sedative doses used during CMV and IMV.
2. *IMV prevents respiratory alkalosis.* In a prospective study comparing IMV and CMV in patients with acute respiratory failure, CMV resulted in a mean pH of 7.49 and a Pa_{CO_2} of 32.3 mm Hg, whereas IMV resulted in a mean pH of 7.44 and a Pa_{CO_2} of 39.9 mm Hg.⁷ The CMV rate, however, was set to suppress spontaneous respiratory effort, whereas, during IMV, the rate was adjusted downward continuously as long as pH remained greater than 7.30. Hence the result in favor of IMV is to be expected. When a normal pH is achieved at a low IMV rate, subsequent data suggested that it was at the expense of increased work of breathing.^{42,74} Moreover, some patients on mechanical ventilation who are allowed to set their own Pa_{CO_2} do not always choose a normal level (e.g., patients with brain injury). These patients will have persistent respiratory alkalosis regardless of ventilation mode.^{77,96}
3. *IMV improves intrapulmonary gas distribution.* This hypothesis is based on the premise that in the supine position, spontaneous breathing causes inspired gas to be distributed preferentially to dependent lung regions because the dependent diaphragm, which is displaced cephalad, operates at an improved mechanical advantage.⁹⁷ Conversely, during mechanical ventilation, ventilation is distributed preferentially to nondependent lung regions because the diaphragm is not used, and the nondependent lung and chest wall are more compliant.⁹⁸ Therefore, IMV’s combination of spontaneous and mechanical breaths theoretically should result in a better matching of ventilation and perfusion.⁹⁹ A comparison study evaluating IMV and CMV ventilation–perfusion distribution with the inert-gas elimination technique¹⁰⁰ was performed in stable patients recovering from major abdominal aortic surgery.¹⁰¹ During CMV, V_T was set at 14 to 16 mL/kg with a set rate of 8 to 10 breaths/min. During SIMV, the rate was set at 50% of the CMV rate (4 to 5 breaths/min); a pressure support of 5 to 8 cm H₂O was added to the intervening spontaneous breaths to overcome the ventilator circuit and endotracheal tube resistance.¹⁰² Each ventilator mode was maintained for 45 minutes. Compared with CMV, physiologic dead space increased with SIMV (22.0% vs. 26.8%, respectively; $p < 0.05$) and was associated with a significant increase in $\dot{V}_E$, resulting in a similar Pa_{CO_2} . The SIMV perfusion distributions remained unaltered. This study, using the inert-gas elimination technique, shows that IMV does not improve ventilation–perfusion distributions.
4. *IMV lowers mean airway pressure, benefiting cardiac output and preventing barotrauma.* Because IMV intersperses spontaneous breaths between mechanical breaths, mean airway pressure averaged over time is lower than with CMV. The lower mean airway pressure results in maintenance of cardiac output. Several studies have shown a higher cardiac output with IMV than with CMV.^{90,91,103} The interactions between intrathoracic pressure associated with mechanical ventilation and cardiac performance, however, are quite complex.¹⁰⁴ Right-ventricular and left-

ventricular interaction, direct pressure on the heart, and changes in systemic and pulmonary venous return all play a role. The net effect of this interaction depends on left-ventricular filling pressure and reserve. Mathru et al¹⁰⁵ compared the effect of CMV, IMV with 5 cm H₂O PEEP (IMV-5PEEP), and IMV with 0 cm H₂O PEEP (IMV-0PEEP) in two groups of patients following aortocoronary bypass surgery. V_T and ventilator rate were adjusted to achieve a PaCO₂ of between 35 and 40 mm Hg. One group had normal left-ventricular function. The second group had decreased left-ventricular reserve: left-ventricular diastolic pressure of greater than 16 mm Hg and ejection fraction of less than 0.6. In the first group, IMV-0PEEP resulted in an increase in cardiac output of 27% compared with CMV. In the second group, however, IMV resulted in a significant decrease in cardiac output (19%) compared with CMV. IMV-5PEEP affected cardiac output similarly to CMV. These data indicate that when compared with CMV, IMV improves cardiac output in patients with normal left-ventricular function or hypovolemia, but it may be harmful in patients with poor left-ventricular reserve.

The frequency of barotrauma with IMV and CMV was compared in a retrospective study of 292 postoperative and nonsurgical patients who received mechanical ventilation for 24 hours or more.¹⁰⁶ The ventilator V_T was set at 12 to 15 mL/kg. The IMV rate was set at 6 breaths/min or lower, provided that normocapnia was maintained. If hypercapnia developed, it was treated by increasing V_T to 15 mL/kg and, if necessary, by increasing the IMV rate to 8 breaths/min. No sedation or muscle relaxants were used. In the CMV group, the rate was set at 12 breaths/min. Hypercapnia was corrected by increasing V_T. The patients were sedated and paralyzed. Compared with the CMV group, patients receiving IMV had a significantly higher peak airway pressure (34 vs. 51 cm H₂O, respectively) and PEEP/CPAP (15 vs. 27 cm H₂O, respectively). Yet the barotrauma was 22% in the CMV group compared with 7% in the IMV group. The authors speculated that the less frequent barotrauma with IMV was related to the smaller number of mechanical breaths with large V_T. Mean transpulmonary pressure, which may be responsible for the barotrauma, was not measured in either group.

5. *IMV decreases oxygen consumption.* Downs et al⁷ found that $\dot{V}_{O_2}$ was lower both during mechanical ventilation and 15 minutes after its discontinuation in patients ventilated with IMV versus CMV. The authors speculated that the higher $\dot{V}_{O_2}$ on CMV could be ascribed to respiratory alkalosis^{107,108} and abrupt withdrawal from mechanical ventilation rather than reflecting metabolic work of breathing. In contrast, Wolff et al¹⁰³ showed that CO₂ production ($\dot{V}_{O_2}$) tended to be lower with CMV than with IMV. Because the respiratory quotient was similar with both ventilation modes, it is reasonable to assume that $\dot{V}_{O_2}$ during CMV was lower than that during IMV. In this study, PaCO₂ was 39 and 44 mm Hg on CMV and IMV, respectively, suggesting that pH was within the normal range.¹⁰³ This observation suggests that, in the absence of alkalemia, $\dot{V}_{O_2}$ is lower on CMV than on IMV.
6. *IMV prevents muscle atrophy and discoordination.* Maintaining spontaneous breathing activity during IMV has been proposed to achieve respiratory muscle conditioning and preserve respiratory muscle function.⁵ The periodic hyperinflation also may reinforce coordinated breathing.^{5,109} During CMV, muscle atrophy^{84,85,110,111} and myofibrillar damage,^{82,112,113} which account for ventilator-induced diaphragmatic dysfunction, have been demonstrated in both experimental animals and humans.^{84,85} Like in experimental animals, in the human diaphragm the atrophic protein kinase B–forkhead Box-O (AKT-FOXO) signaling plays an important role in activating the ubiquitin-proteasome pathway.¹¹⁴ Overexpressions of the atrophic genes, atrogin-1 and muscle ring finger-1

(MuRF-1) of the ubiquitin-proteasome pathway, are responsible for myofibril degradation and atrophy with CMV-induced diaphragm muscle inactivity.¹¹⁴ Conversely, maintaining diaphragmatic activation with ACV attenuates expression of atrogin-1 (muscle atrophy F-box [MAF-box]; see [Chapter 43](#)).⁸³ With IMV, diaphragmatic electrical activity of both mandatory and spontaneous breaths persists. Thus, it is possible that IMV prevents respiratory muscle atrophy. This hypothesis remains to be tested.

Respiratory muscle discoordination with IMV and CMV has not been compared. IMV's efficacy in counteracting respiratory muscle discoordination was demonstrated by Andersen et al¹⁰⁹ in a study of twenty-eight patients during discontinuation of mechanical ventilation. The mandatory breaths were increased gradually to 75% of the patient's $\dot{V}_E$. In contrast, Gibbons et al¹¹⁵ failed to show similar results when IMV was applied to six patients receiving prolonged mechanical ventilation. The lowest spontaneous breathing rate was 29 breaths/min at the lowest IMV rate of 10 breaths/min. Gas exchange was adequate. Five of the six patients manifested breathing discoordination in the form of either ribcage or abdominal paradox. In the patients whose diaphragmatic electrical activity was measured, an electromyographic fatigue pattern also was observed. In this study, IMV was insufficient to reduce respiratory muscle workload, as reflected by the relatively high spontaneous breathing rate. IMV's efficacy in counteracting respiratory muscle discoordination appears to depend primarily on the extent of respiratory muscle unloading.

7. *IMV improves renal function.* Steinhoff et al¹¹⁶ studied the effect of CMV and IMV on renal function in patients with acute respiratory failure. With CMV, the ventilator rate was set to suppress inspiratory efforts, which averaged 10 to 16 breaths/min. IMV was set at 4 to 10 breaths/min. PEEP was maintained constant during both CMV and IMV. With CMV, urinary flow and creatinine and osmolal clearance decreased, with a net effect of water retention, in comparison with IMV. Impaired renal function during CMV was attributed to the increased intrathoracic pressure, which caused stimulation of atrial stretch receptors and release of antidiuretic hormone. A more important factor may be that increased intrathoracic pressure during CMV decreases venous return, and the consequent decrease in cardiac output produces a decrease in renal plasma flow. Hence the effects of CMV and IMV on renal function are related directly to their respective effects on cardiac function.

Intermittent Mandatory Ventilation and Assist-Control Ventilation

Comparison of IMV with ACV is more appropriate than comparison of IMV with CMV because both IMV and ACV provide partial ventilator support. Few studies have compared IMV with ACV. As discussed below, these studies primarily concern effects on cardiac output,^{117,118} $\dot{V}_{O_2}$,^{117–119} and respiratory alkalosis.^{77,117,120} (Because IMV's effects on work of breathing were discussed extensively earlier, comparison of IMV with ACV is limited to its application in the neonate.^{121,122})

1. *Effect on cardiac output.* Groeger et al¹¹⁷ studied the effect of SIMV and ACV in forty patients with acute respiratory failure of various etiologies other than chronic obstructive pulmonary disease. SIMV, at a set V_T of 10 to 15 mL/kg, was the initial ventilation mode in all patients, and the mandatory rate was adjusted to the minimum required to maintain a normal pH and P_{aCO_2} . When the combined mandatory and spontaneous breathing rates were greater than 35 breaths/min, the IMV rate was increased to ensure patient comfort. The ventilator mode was then switched to ACV, and V_T , PEEP, inspiratory flow rate, and inspired oxygen fraction (F_{IO_2}) were held constant. After 30 minutes on SIMV or ACV, hemodynamic variables were measured. At this point, mean IMV rate was 7 breaths/min, with a total rate of 34 breaths/min, whereas the ACV rate was 15 breaths/min. Cardiac output, measured by thermodilution, was 6% higher

with SIMV than with ACV. Likewise, studying twelve patients recovering from acute respiratory failure of various etiologies, Sternberg and Sahebjam¹¹⁸ demonstrated that cardiac index was significantly higher with SIMV than with ACV (3.6 vs. 3.3 L/min/m²). (The investigators also compared SIMV with PSV, which is discussed in the section [Intermittent Mandatory Ventilation and Pressure-Support Ventilation](#).) The average V_T with ACV was 715 mL, whereas with SIMV it was 491 mL. The SIMV mandatory breaths were set at 75% of the ACV rate. Although the cardiac output with SIMV was significantly higher than with ACV, the changes fall within the variability of the thermodilution technique.¹²³ Despite the limited differences in cardiac output with IMV and ACV, IMV may be helpful in patients who demonstrate significant hemodynamic deterioration during ACV. In the original description of intrinsic PEEP (PEEPi),¹²⁴ two patients with chronic obstructive pulmonary disease developed hemodynamic compromise secondary to significant PEEPi while receiving ACV. The institution of IMV and fluid repletion produced an improvement, although cardiac output was not measured directly during either IMV or ACV. The conflicting data concerning the effects of ACV and IMV on cardiac output underscore the complex interaction between intrathoracic pressure and cardiac function.

2. *Effect on oxygen consumption.* In the above-mentioned study by Groeger et al,¹¹⁷ $\dot{V}_{O_2}$ was comparable for both pressure-triggered SIMV and ACV. When the patients were grouped according to the ratio between $\dot{V}_E$ achieved by the mandatory breaths and total $\dot{V}_E$, however, the mean $\dot{V}_{O_2}$ for patients with a ratio of less than 0.5 was significantly higher during SIMV than during ACV (320 vs. 296 mL/min/m², respectively; $p \leq 0.05$). Conversely, in the study of Sternberg and Sebahjani,¹¹⁸ when the mandatory breaths during SIMV were set at 75% of the ACV rate, $\dot{V}_{O_2}$ was unaltered during SIMV despite total frequency being higher with SIMV than with ACV (20 breaths/min vs. 12 breaths/min). In healthy subjects breathing via a mouthpiece on SIMV or ACV, $\dot{V}_{O_2}$ (measured with a metabolic cart) also was similar with the two modes.¹¹⁹ With SIMV, both the V_T and mandatory-breath rate were equivalent to those of the ACV. These studies^{117–119} demonstrate that compared with ACV, the degree of machine assistance during IMV determines $\dot{V}_{O_2}$.
3. *Effect on respiratory alkalosis.* Three prospective studies^{77,117,120} comparing IMV and ACV showed a significantly lower pH and higher Pa_{CO_2} during IMV than during ACV ([Table 7-1](#)). Groeger et al¹¹⁷ suggested that the higher Pa_{CO_2} during IMV was related to an increased dead-space-to-tidal-volume ratio, because $\dot{V}_{CO_2}$ and $\dot{V}_E$ were similar for both IMV and ACV. Conversely, Hudson et al⁷⁷ showed that the higher Pa_{CO_2} during IMV came at the expense of a high patient workload given elevated CO_2 and an unchanged alveolar ventilation level. Regardless of the mechanisms of the elevated Pa_{CO_2} during IMV, all three groups of investigators concluded that the decrease in pH and increase in Pa_{CO_2} were minimal and of questionable clinical significance. Furthermore, in studies where IMV was compared with ACV in a subgroup of patients with preexisting respiratory alkalosis, respiratory alkalosis persisted during IMV.^{77,120}
4. *Effects on work of breathing in the neonate.* Kapasi et al¹²² undertook a comparison of the effects of pressure-limited ACV, SIMV, and IMV on work of breathing and inspiratory effort of clinically stable neonates with respiratory distress syndrome. The mandatory breath rate with both SIMV and IMV was set the same as that of the ACV (range: 14 to 25 breaths/min). Average total respiratory rate was not significantly different among the modes (IMV, 56.3 breaths/min; SIMV, 58.3 breaths/min; ACV, 58.8 breaths/min). The work of breathing was estimated using the esophageal pressure,

calculated using the Campbell diagram.^{125,126} Both work of breathing and inspiratory effort were least with ACV and highest with IMV; SIMV had values between ACV and IMV. Patient-ventilator asynchrony occurred only with IMV. Jarreau et al¹²¹ reported a similar result when comparing IMV, ACV with inspiratory pressure set at 10 to 15 cm H₂O, and spontaneous breathing on CPAP. Work of breathing with IMV was similar to that with CPAP: 0.81 versus 0.90 J/L. Work of breathing fell significantly only during ACV to 0.48 to 0.47 J/L at inspiratory pressures of 10 and 15 cm H₂O, respectively. Thus, in neonates, patient-triggered ACV provides better patient-ventilator synchrony and unloading of workload than does continuous-flow IMV.

Table 7-1: Effect of IMV versus ACV on pH and PaCO₂

	pH		PaCO ₂ (mm Hg)		Rate (Breaths Per Minute)		
Reference No.	IMV	ACV	IMV	ACV	IMV	ACV	<i>n</i>
63	7.41 ± 0.06 ^a	7.45 ± 0.06	43.0 ± 6.3 ^a	38.0 ± 6.3	7.1 (33.6)	15.1	40
102	7.48 ± 0.05 ^b	7.51 ± 0.04	29.7 ± 6.1	28.6 ± 4.9	1/2 ACV rate	NA	26
105	7.42 ± 0.08 ^a	7.45 ± 0.04	40.7 ± 7.6 ^a	37.9 ± 6.7	4 (21)	15.0	18
Patients with Preexisting Respiratory Alkalosis							
63	7.49 ± 0.03	7.49 ± 0.03	27.4 ± 6.3	29.1 ± 4.7			17
105	7.46 ± 0.07 ^b	7.49 ± 0.03	37.8 ± 7.4	35.7 ± 6.7			12

^a*p* < 0.01.

^b*p* < 0.05.

Note: Values are mean ± SD; only mean rate is shown. Number in parenthesis denotes total mandatory and spontaneous breath rates. Tidal volume was maintained the same for the mandatory breath with both IMV and ACV.

Abbreviations: IMV, intermittent mandatory ventilation; ACV, assist-control ventilation; NA, not available.

Sources: Adapted, with permission, from Hudson et al,⁷⁷ Groeger et al,¹¹⁷ and Culpepper et al.¹²⁰

In summary, the limited number of studies suggests minimal differences between the effects of high levels of IMV and ACV on cardiac output, $\dot{V}_{O_2}$, and respiratory alkalosis. Because of lower mean airway and intrapleural pressures, IMV should help to improve cardiovascular function in patients who exhibit hemodynamic compromise during ACV. In neonates, ACV is more effective in unloading the work of breathing and providing synchrony than IMV.

Intermittent Mandatory Ventilation and Pressure-Support Ventilation

As with IMV, PSV provides the patient with some autonomy to alter breathing patterns in response to ventilatory demand. PSV is a form of ventilatory support in which the patient's inspiratory effort is assisted by the ventilator up to a preset inspiratory pressure level and remains at that level until a fixed^{127,128} or operator-adjustable¹²⁹ ventilator cycle-off algorithm is activated. Unlike IMV, in which the number of mandatory breaths is fixed, PSV assists every breath, and the ventilator contribution to total workload is variable. Because the set inspiratory pressure is fixed with PSV, when patient ventilatory demand increases, inspiratory effort may exceed the ventilator contribution to total workload (see [Chapter 8](#)).^{130,131} PSV can be added to IMV to unload inspiratory muscle work during the spontaneous breathing cycles. The addition of a small amount of PSV (5 cm H₂O) to pressure-triggered SIMV is adequate to overcome the lack of flow delivery observed with some pressure-triggered SIMV systems.⁴¹ Higher levels of PSV not only help in overcoming the ventilator circuit and endotracheal tube resistance^{132,133} but also augment V_T and unload the elastic work of the spontaneous breaths.^{134,135} In one study of a small number of patients with acute respiratory failure,¹³⁶ PSV application at levels of up to 30 cm H₂O during SIMV (IMV rate of 6 to 10 breaths/min) and PEEP of 3 to 13 cm H₂O did not result in cardiovascular compromise. In clinically stable preterm infants, application of pressure support to the intervening spontaneous breaths to deliver V_T of 5 to 8 mL/kg stabilized breathing pattern,¹³⁷ and prevented an increase in inspiratory effort when SIMV rate was reduced.¹³⁸ Comparison between IMV and PSV is discussed pertaining to cardiac output,¹¹⁸ ventilation-perfusion distribution,¹³⁹ and unloading of patient effort.⁸⁴

1. *Effects on cardiac output.* Hemodynamics during SIMV and PSV were studied in critically ill patients by initially applying ACV to the patients.¹¹⁸ The V_T of the mandatory breaths with SIMV then was set the same as that of ACV, at a rate of 75% of the ACV rate. With PSV, the inspiratory pressure was set to produce a V_T similar to that of ACV (average pressure of 21 cm H₂O). Cardiac output (measured by thermodilution), oxygen transport, and $\dot{V}_{O_2}$ were the same for both ventilation modes. As a primary means of ventilatory support, both SIMV and PSV have comparable effects on hemodynamics.
2. *Effects on ventilation-perfusion.* Valentine et al¹³⁹ studied the effect of SIMV, PSV, and airway pressure-release ventilation (see in the section [Intermittent Mandatory Ventilation and Airway Pressure-Release Ventilation](#)) on ventilation-perfusion distribution in post-cardiac surgery patients who were ready to be weaned; this section discusses only the comparison between SIMV and PSV. SIMV was the initial ventilatory support mode. The IMV rate was adjusted to maintain a pH of greater than 7.35. With PSV, pressure was titrated to achieve a mean end-tidal P_{CO_2} of 40 mm Hg. Ventilation-perfusion distribution was assessed using the inert-gas elimination technique.¹⁰⁰ The dispersion of ventilation-perfusion ratios, calculated as the logarithmic standard deviation of perfusion ($\log SD \dot{Q}$) and ventilation ($\log SD \dot{V}$), was similar for SIMV and PSV. Right-to-left intrapulmonary shunt and fractional dead-space ventilation did not differ significantly. [Table 7-2](#) shows the effects on arterial blood gases, respiratory mechanics, and $\dot{V}_{O_2}$. Differences in arterial blood gases were of questionable clinical significance. Peak airway pressure was significantly higher with SIMV, but mean transpulmonary pressure was comparable. As sole ventilator support, SIMV and PSV provide comparable and adequate gas exchange in postoperative patients who were ready to be weaned. No study has yet compared the efficacy of IMV and PSV as a primary means of ventilator support during acute respiratory failure.
3. *Effects on patient effort.* Leung et al⁴³ carried out a head-to-head comparison of the efficacy of SIMV and PSV in unloading inspiratory effort at various levels of assistance. The rate of change in inspiratory effort with increasing

assistance levels, estimated as pressure-time product per minute, did not differ between the two modes. Unloading efficacy, however, differed according to the level of assistance. From 0% to 60% of maximum, the decrease in pressure-time product per minute was greater with PSV than with SIMV. At a higher assistance level, the converse was observed (Fig. 7-11).¹⁴⁰ Frequency decreased linearly with increase in PSV. With SIMV, frequency changed little until a high assistance level was provided (Fig. 7-11).¹⁴⁰ Thus, when a high assistance level is needed, both SIMV and PSV provide comparable assistance. At low to medium assistance levels, however, a greater decrease in patient effort with PSV makes it more useful clinically than SIMV. Leung et al⁴³ also assessed the patient's wasted efforts or nontriggered attempts during both SIMV and PSV. The number of nontriggered attempts is proportional to the assistance level and highest at 100% of machine assistance (29% with SIMV and 26% with PSV). The breaths preceding the nontriggered attempts had a shorter total duration and expiratory time, higher V_T , and higher dynamic PEEPi than did the breaths preceding triggered breaths. This observation suggests that nontriggered attempts resulted from an inspiratory effort that was insufficient to overcome the elevated recoil pressure associated with dynamic hyperinflation. Thus, increasing the ventilator assistance level decreases inspiratory muscle effort but also increases ineffective triggering.^{43,141} Ineffective triggering is associated with prolonged duration of mechanical ventilation.¹⁴²

Table 7-2: Gas Exchange, Mechanics, and Oxygen Consumption during SIMV and PSV

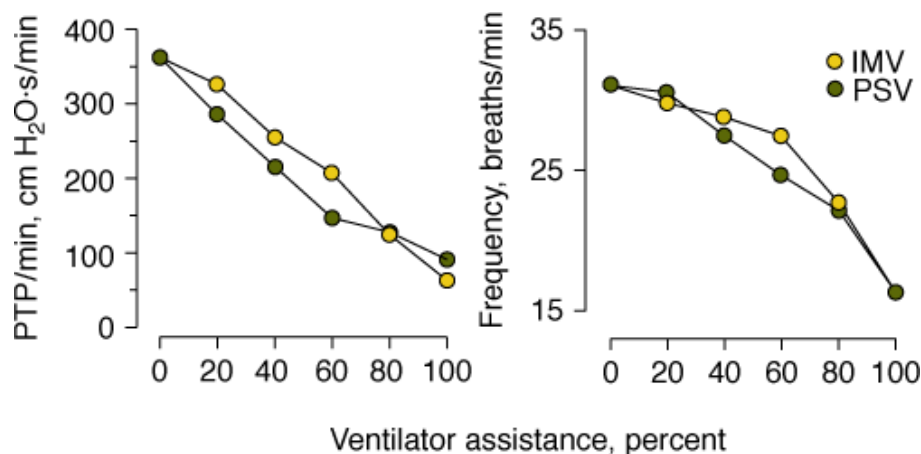
	SIMV	PSV
FI _{O2}	0.44 ± 0.11	0.44 ± 0.11
pH	7.41 ± 0.02	7.36 ± 0.02 ^a
PaCO ₂ , mm Hg	33.0 ± 2.0	39.0 ± 2.0 ^a
PaCO ₂ , mm Hg	102.0 ± 7.0	98.0 ± 8.0
Peak Paw, cm H ₂ O	32.8 ± 1.3	19.4 ± 2.1 ^a
Mean Paw, cm H ₂ O	9.6 ± 1.1	8.4 ± 1.0
Ppl, cm H ₂ O	3.8 ± 1.0	3.8 ± 1.1
Ptp, cm H ₂ O	5.8 ± 0.6	4.6 ± 0.5
$\dot{V}_E$, liters/min	9.4 ± 0.6	9.0 ± 0.5
fS, breaths per minute	3.4 ± 1.8	
fM, breaths per minute	8.4 ± 0.4	15.8 ± 0.9 ^a
V _T S, liters	0.08 ± 0.07	
V _T M, liters	1.03 ± 0.03	0.58 ± 0.03 ^a
$\dot{V}_{O_2}$, ml/min	269 ± 13	268 ± 14

^a $p < 0.05$.

Note: Values are mean ± SE; $n \geq 9$.

Abbreviations: SIMV, synchronous intermittent mandatory ventilation; PSV, pressure-support ventilation; FI_{O2}, inspired oxygen fraction. Paw, airway pressure; Ppl, pleural pressure; Ptp, transpulmonary pressure; $\dot{V}_E$, total minute ventilation; fS, frequency of spontaneous breaths during SIMV; fM, frequency of mandatory breaths or pressure support assisted breaths; V_TS, tidal volume of spontaneous breath during SIMV; V_TM, tidal volume of mandatory breath or pressure-support assisted breath; $\dot{V}_{O_2}$, oxygen consumption.

Sources: Adapted, with permission, from Valentine et al. ¹³⁹



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Changes in PTP per minute (*left panel*) and frequency (*right panel*) as intermittent mandatory ventilation (IMV) and pressure-support ventilation (PSV) were increased progressively. PSV of 100% represents the level necessary to achieve a V_T equivalent to that during ACV (10 mL/kg); IMV 100% is the same ventilator rate and V_T as during ACV. (Adapted, with permission, from Leung P, Jubran A, Tobin MJ. Comparison of assisted ventilator modes on triggering, patient effort, and dyspnea. *Am J Respir Crit Care Med*. 1997;155:1940–1948; and Tobin MJ, Jubran A, Laghi F. Patient-ventilator interaction. *Am J Respir Crit Care Med*. 2001;163:1059–1063.)

Intermittent Mandatory Ventilation and Airway Pressure-Release Ventilation

Airway pressure release ventilation (APRV) consists essentially of two CPAP levels with a transient decrease or “release” of airway pressure from a higher CPAP to a lower CPAP for a set release time. Spontaneous breathing is allowed to occur between airway pressure releases.^{128,143–145} The pressure gradient between the two CPAP levels and the frequency of releases determine the level of ventilator support (see [Chapter 11](#)). The original version of APRV used a short release time, and in the absence of spontaneous breathing, APRV resembled pressure-controlled inverse-ratio ventilation.¹⁴³ The modified version, in which the inspiration-to-expiration (I:E) ratio is adjustable, is termed *biphasic intermittent positive airway pressure*.¹⁴⁵ The primary indication for APRV is the provision of ventilation during an oxygenation crisis.¹⁴³ Although the indications and operating principles of IMV and APRV differ significantly, both modes allow the patient to breathe spontaneously between machine-cycled breaths.^{143,144} Comparison of IMV with APRV is discussed with regard to its efficacy as a primary means of ventilator support in patients with acute respiratory distress syndrome (ARDS)¹⁴⁶ in relation to gas exchange,^{139,146–149} hemodynamics,^{146,148,150} lung recruitment,¹⁵¹ breathing comfort,^{148,152} intubation duration, and sedative use.^{146,153}

Varpula et al¹⁴⁶ undertook a randomized, controlled comparison of pressure-limited SIMV (with 10 cm H₂O of pressure support added to the unassisted breaths) and APRV in patients with ARDS. The primary end point was the number of ventilator-free days from the time of randomization to day 28. The secondary end points were the effect on gas exchange, hemodynamics, and sedative use. PEEP was set slightly above the lower inflection point on the pressure-volume curve obtained during paralysis (or if not detected, at 10 cm H₂O). The upper inflection point was never exceeded (or if not detected, the inspiratory pressure was set at less than 35 cm H₂O). The pressure-release frequency with APRV and the machine assistance rate with SIMV were similar at 12 breaths/min. The I:E ratio with APRV was set at 4:1; with SIMV, it was set at 1:2 and adjusted to 2:1. Ventilator-free days, gas exchange, hemodynamics, and sedative dosage were comparable

with the two modes, although average inspiratory pressure was significantly lower with APRV than with SIMV (25.9 vs. 28.6 cm H₂O). Moderate hypercapnia developed during the first study day in both groups but returned toward normal after 4 days. The effects of SIMV and APRV on gas exchange and hemodynamics appear comparable in patients with lung injury^{147,148} or after cardiac surgery.¹³⁹ In contrast to earlier studies, recently, one trial in patients with ARDS demonstrated the superiority of APRV over SIMV in terms of gas exchange and mortality.¹⁴⁹ Mortality was 31% with APRV compared to 59% with SIMV. Unfortunately, this study was a retrospective trial conducted over 6 years with SIMV applied in the first 3 years followed thereafter with APRV; consequently, ventilator settings were not well matched.¹⁴⁹ Application of short-term APRV also proved superior to SIMV in patients who underwent coronary artery bypass surgery.¹⁵⁰ Intubation duration was approximately 4 hours shorter with APRV than with SIMV (mean: 10.1 vs. 14.7 hours). Analgesic and sedative dosage also was less with APRV than with SIMV. This study, however, was a nonrandomized, open clinical trial.

In a recent, prospective, randomized trial in acute lung injury using similar ventilator settings as above,¹⁴⁶ Varpula et al¹⁵¹ compared the effects APRV ($n = 13$) and SIMV plus PSV ($n = 10$) in recruiting nonaerated lungs. The investigators hypothesized that a strategy employing spontaneous breathing with active diaphragmatic contractions, as with APRV, is associated with better aeration of the dependent lung region.⁹⁸ The degree of lung recruitment was estimated using computer-assisted tomographic densitometry technique at the beginning of randomization, and after 7 days of the assigned ventilator mode. On day 7, the median peak Paw was significantly lower with APRV than with SIMV: 24.0 versus 30.4 cm H₂O. The computer-assisted tomographic number (in Hounsfield units) was similar with both APRV and SIMV at both the carinal and dependent diaphragmatic level. Thus, APRV did not appear to improve aeration of the consolidated lung parenchyma.

Breathing comfort during SIMV and APRV were evaluated in inexperienced healthy subjects breathing via a mouthpiece and compared with PSV. SIMV was set at 8 breaths/min, V_T at 5 mL/kg, and PEEP at 5 cm H₂O. The low and high PEEP levels with APRV were set at 5 and 10 cm H₂O, the release rate was 8 breaths/min, and the I:E ratio was 1:2. PSV was set at 10 cm H₂O with PEEP of 5 cm H₂O. Breathing comfort was measured with a visual analog scale (0 to 10 cm). PSV achieved the greatest comfort, SIMV the worst, and APRV fell between PSV and SIMV (2.03, 5.38, and 4.12 cm, respectively). Unfortunately, flow-limited SIMV was employed rather than pressure-limited SIMV (both APRV and PSV are pressure-limited ventilation modes).

Intermittent Mandatory Ventilation and Proportional-Assist Ventilation, Adaptive-Support Ventilation, Pressure-Regulated Volume Control Ventilation, and Volume Support Ventilation

Proportional-Assist Ventilation

Proportional-assist ventilation (PAV) is a mode in which the ventilator instantaneously generates pressure in proportion to the patient's effort (see Chapter 12).¹⁵⁴ The ventilator amplifies the patient's inspiratory effort without any preselected target volume or pressure. A randomized crossover comparison of continuous-flow IMV versus PAV was conducted in thirty-six preterm infants.¹⁵⁵ The inspiratory pressure with IMV was set to deliver a V_T of 4 to 6 mL/kg. The IMV rate was not reported. With PAV, the volume-assist gain was adjusted to decrease lung elastance to its normal value, whereas the flow-assist gain was set at -20 cm H₂O/L/s. Each mode was applied for 45 minutes. With PAV, peak and mean airway pressures and transpulmonary pressure were significantly lower, frequency was higher, and V_T was comparable to

that with IMV, resulting in a higher $\dot{V}_E$. PaCO_2 , however, remained the same as with IMV, PaO_2 and was higher. There were no significant differences in the number of apneic or hypoxemic episodes. The lower transpulmonary pressure with PAV might help to prevent lung injury during prolonged mechanical ventilation. The results in preterm infants were similar to those in an earlier short-term trial in a few adult patients.¹⁵⁶

Adaptive-Support Ventilation

Adaptive-support ventilation (ASV) is based on the work by Otis et al¹⁵⁷ and Mead et al,¹⁵⁸ who demonstrated that any given level of alveolar ventilation has an optimal respiratory frequency that is least costly in terms of respiratory work: the frequency at which the respiratory muscles develop the least-average force or tension.¹⁵⁸ ASV is a mode that can alternate between pressure control and pressure support, relying on closed-loop regulation of ventilator settings that respond to changes in respiratory system mechanics and spontaneous breathing efforts (see Chapter 15). ASV adjusts inspiratory pressure, I:E ratio, and mandatory rate to maintain the target minute ventilation and respiratory rate within a frame designed to avoid both rapid, shallow breathing and excessive inflation volumes.¹⁵⁹ Tassaux et al¹⁶⁰ compared the short-term effects of ASV versus SIMV plus pressure support on patient-ventilator interaction in patients ready to be weaned from mechanical ventilation. ASV achieved a $\dot{V}_E$ similar to that of SIMV. Central neural drive, however, estimated as $\text{P}_{0.1}$, and sternocleidomastoid electrical activity, measured with surface electrodes, were reduced markedly. Thus, ASV provided comparable total $\dot{V}_E$ but with a significantly decreased inspiratory load than did SIMV plus pressure support.

Pressure-Regulated Volume-Controlled Ventilation

Pressure-regulated volume-controlled ventilation (PRVCV) is a dual-control, breath-to-breath mode. PRVCV has both the benefits of pressure-controlled ventilation, with a constant V_T , and automatic weaning from pressure limit as patient compliance improves and/or patient effort increases (see Chapter 15). Four prospective, randomized trials have compared the efficacy of PRVCV and IMV with different primary goals in preterm infants.^{161–164} Two trials were short-term in stable preterm infants during the weaning phase,^{161,162} and two trials during acute respiratory failure.^{163,164} In the stable infants, SIMV was compared with PRVCV on the effects of respiratory mechanics and gas exchange.^{161,162} The SIMV mandatory breaths consisted of volume guarantee or pressure-regulated volume-controlled breaths. Abubakar and Keszler¹⁶¹ concluded that SIMV resulted in higher work of breathing than PRVCV on the basis of a high respiratory rate, although work of breathing was not directly measured. Scopesi et al¹⁶² demonstrated that SIMV was associated with lower mean Paw and higher variability in V_T than observed with PRVCV. From those studies^{161,162} it is not clear whether PRVCV offers advantage over SIMV because the sample size was small (ten to twelve patients), and trial duration was short (20 minutes to 2 hours).

In infants with respiratory distress syndrome, Piotrowski et al¹⁶³ studied infants younger than 3 days old whose birthweight was less than 2500 g. Thirty infants received continuous-flow IMV, and twenty-seven received PRVCV; the average Paw were 18.6 and 16.2 cm H_2O , respectively. The IMV rate was selected by the clinician. With PRVCV, the V_T was set at 5 to 6 mL/kg. The primary end point was duration of mechanical ventilation and incidence of bronchopulmonary dysplasia. The secondary end point was complications from mechanical ventilation, consisting of air leaks, intraventricular hemorrhage, and hemodynamic instability. PRVCV did not decrease duration of mechanical ventilation or incidence of bronchopulmonary dysplasia, although it decreased the incidence of high-grade intraventricular hemorrhage (11% vs. 35%). The benefit may have occurred in part because PRVCV delivers a stable volume. In preterm

newborns, large fluctuations in intrathoracic and arterial pressures cause variations in cerebral blood flow velocity that is a risk factor for intraventricular hemorrhage.¹⁶⁵ In the second trial, D'Angio et al¹⁶⁴ enrolled a large number of low-birthweight infants (500 to 1249 g) who were younger than 6 hours of age. Infants were assigned to pressure-limited SIMV ($n = 108$) or PRVCV ($n = 104$) until extubation, death, or meeting predetermined failure criteria on the assigned mode. With SIMV, no pressure support was applied to the spontaneous breaths. Average Paw measured at 6 hours and 12 hours after SIMV application were 15.0 and 14.0 cm H₂O, respectively, and after PRVCV were 13.8 and 12.7 cm H₂O, respectively. The average V_T for SIMV at 6 hours and 12 hours was 18.4 and 17.8 mL/kg, respectively; whereas V_T with PRVCV was 16.0 and 14.1 mL/kg, respectively. The average set ventilator rate with SIMV at 6 hours and 12 hours was 35 and 30 breaths/min, respectively; with PRVCV, the rate was 40 breaths/min at both 6 hours and 12 hours. Differences in measured ventilator variables between SIMV and PRVCV were not significant. The cumulative percentage of infants alive and extubated at 36 weeks (84%) was similar for both SIMV and PRVCV (88 out of 105 infants for SIMV; 87 out of 104 infants for PRVCV). The percentage of infants who failed the assigned mode tended to be less with PRVCV than with SIMV, 20% versus 33%. Complications associated with prematurity and mechanical ventilation such as bronchopulmonary dysplasia, intraventricular hemorrhage, or retinopathy of prematurity were similar. Interestingly, with both SIMV and PRVCV, V_T was remarkably higher than that recommended with a protective ventilatory strategy.¹⁶⁶ Yet, when ventilator settings are nearly matched, application of SIMV or PRVCV appears to have a similar impact on survival, extubation rate, or complications rate in preterm infants with acute respiratory failure.

Volume-Support Ventilation

Volume-support ventilation (VSV), or PSV combined with volume guarantee, is PSV that uses V_T as feedback control for continuously adjusting the pressure-support level. All breaths are patient-triggered, pressure-limited, and flow-cycled.⁵² Prospective comparisons between VSV and SIMV in preterm infants were conducted during the acute¹⁶⁷ and stable phase of respiratory failure.^{162,168} Nafday et al¹⁶⁷ randomized preterm infants following surfactant treatment for respiratory distress syndrome into VSV ($n = 16$) or pressure-limited SIMV ($n = 18$) applied for 24 hours. V_T with VSV was set at 5 mL/kg. Ventilator settings were adjusted to achieve predetermined arterial blood-gas values. After 24 hours, attending physicians provided ventilation management as clinically indicated. Mean airway pressure decreased over time in both VSV and SIMV groups, although the decrease was faster in the SIMV group. Both groups had similar survival rates at the time of discharge (86% for VSV, and 94% for SIMV), and complications associated with prematurity and mechanical ventilation. That is, VSV did not offer improved clinical outcomes compared with SIMV.

Two prospective, short-term trials of stable preterm infants during the weaning phase compared VSV and pressure-limited SIMV for either 30 minutes¹⁶⁸ or 20 minutes.¹⁶² In the first trial,¹⁶⁸ following either VSV or SIMV for 30 minutes, VSV was applied for 24 hours. The attending physician, however, had discretion to select a ventilation mode according to the infant's condition. VSV was successfully applied for 24 hours in twenty-one of 25 infants (84%), in those infants a significantly longer duration of rhythmic breathing was observed during the 30-minute trial. VSV provided comparable fluctuations in ventilation and oxygenation, but with a lower Paw than with SIMV (15.4 vs. 19.2 cm H₂O, respectively), and shorter T_I (0.29 vs. 0.35 seconds, respectively). In infants with high respiratory drive, however, hyperventilation occurred with VSV, which dictated a switch to SIMV.¹⁶⁸ Consistent with the first trial, the second trial also demonstrated lower Paw with VSV compared with SIMV (11.2 vs. 18.2 cm H₂O, respectively). In summary, VSV, in the short term, did not offer ventilatory advantage over SIMV in acute or stable preterm infants with respiratory distress syndrome, except for a lower Paw in the stable infants.

Intermittent Mandatory Ventilation as a Weaning Method

Intermittent Mandatory Ventilation, Pressure Support, and T Piece

IMV was first used in adult patients as a means of discontinuing mechanical ventilation.⁵ This method was claimed to be more efficient, safer, and more readily accepted by the patient, and it avoided the necessity of setting up a T-piece circuit. Although preceding ventilator support may be with either ACV or IMV, IMV is applied when the patient is ready to be weaned. The number of mandatory breaths is reduced gradually (1 to 3 breaths/min) at 1- to 4-hour intervals, provided that arterial pH remains greater than 7.30⁷ or 7.35,¹⁶⁹ regardless of other physiologic measurements. An IMV rate of zero or close to zero is maintained for several hours, or for as long as 24 hours, before extubating the patient.

With PSV as a weaning method, the pressure level is set initially at a maximum, defined as the level that produces a V_T of 10 to 12 mL/kg; then the pressure support level is reduced according to the patient's respiratory frequency.¹⁷⁰ When the pressure support level reaches 5 cm H₂O, extubation is considered.

With a T piece, once the patient meets predefined weaning criteria, the patient is placed on a T-piece circuit.¹⁷¹ Progressively longer intervals of spontaneous breathing through a T piece are alternated with ACV. Extubation is considered when the patient can sustain breathing through a T piece for 1 to 2 hours. The early claim for IMV's superiority over a T piece was not based on a controlled study, and subsequent retrospective¹⁷² and prospective studies^{173,174} failed to demonstrate IMV's superiority. Studies comparing IMV and PSV showed either a significantly reduced duration¹⁷⁵ or a tendency for a shorter weaning time¹⁷⁶ with PSV. As PSV grew in popularity, two prospective, randomized, controlled multicenter trials^{177,178} simultaneously compared the three weaning modalities: SIMV, PSV, and T piece. These trials laid to rest IMV's claim to superiority over T piece and PSV.

Brochard et al¹⁷⁷ studied 109 patients who met three of the four defined weaning criteria and had failed a 2-hour T-piece trial. Patients were randomized to SIMV ($n = 43$), PSV ($n = 31$), and T piece ($n = 35$). Weaning failure was defined as continued inability to be weaned after 21 days on the same mode, the need for reintubation after 48 hours of extubation, or intercurrent events (e.g., cardiac ischemia or nosocomial pneumonia) within 72 hours in the selected mode. The initial SIMV rate was set at half the total frequency during ACV or CMV, keeping V_T and flow rates constant (mean initial SIMV rate of 9.5 breaths/min). Once or twice a day, the SIMV rate was decreased by 2 to 4 breaths/min if patients did not demonstrate signs of poor tolerance. When a patient demonstrated poor tolerance, the SIMV rate was increased to its preceding level. When a patient tolerated a SIMV rate of 4 breaths/min or less over 1 day, tracheal extubation was performed. The T-piece method consisted of a gradual lengthening of the periods of disconnection from the ventilator. The initial T-piece trial was set shorter than the initial tolerance duration (mean duration: 38 minutes). The number of T-piece trials depended on the length of disconnection and on nurse availability and varied from three to eight trials per day. The T-piece periods were lengthened incrementally twice a day. Between the T-piece trials, ACV was applied. When the duration of a T-piece trial had reached 2 hours with adequate gas exchange, tracheal extubation was performed. In the patients assigned to PSV, the initial pressure was adjusted until the frequency ranged between 20 and 30 breaths/min. Twice a day, the pressure was decreased by 2 to 4 cm H₂O if the patient did not show any signs of poor tolerance; if tolerance worsened, pressure was increased to its preceding level. When the patient tolerated a PSV level of 8 cm H₂O or less throughout a 24-hour period, tracheal extubation was performed. During the 21-day trial, the probability of being weaned was twice as high with PSV compared with SIMV or T piece. Weaning duration did not differ between SIMV and T piece but was significantly shorter with PSV than with the other two modalities (6 vs. 9 days,

respectively). The number of weaning success patients was significantly larger with PSV (77%) than with SIMV (58%) or T piece (57%) (Table 7-3).

Table 7-3: Trials Comparing Three Weaning Methods: Intermittent T Piece, Pressure-Support Ventilation (PSV), and Synchronized Intermittent Mandatory Ventilation (SIMV)

	Successful Wean [<i>n</i> (%)]		
Trial	T Piece	PSV	SIMV
Brochard et al ¹⁷⁷	20/35 (57)	24/31(77)	25/43 (58)
Esteban et al ¹⁷⁸	27/33 (82)	23/37 (62)	20/29 (69)
	Risk Difference (%) ^a		
	T Piece vs. PSV	T Piece vs. SIMV	PSV vs. SIMV
Brochard et al ¹⁷⁷	-20 (-42, 2)	-1 (-23, 21)	19 (-2, 40)
Esteban et al ¹⁷⁸	14 (-5, 32)	7 (-13, 27)	-7 (-20, 16)
	Time to Extubation (Days)		
	T Piece	PSV	SIMV
Brochard et al ¹⁷⁷	8.5 ± 8.3	5.7 ± 3.7	9.9 ± 8.2
Esteban et al ¹⁷⁸	3(2, 6)	4(2, 12)	5(3, 11)

^aRisk differences expressed as the difference in proportions of successfully weaned subjects between respective weaning methods. Negative numbers imply lower success rates and positive numbers higher success rates. The 95% confidence intervals follow risk differences in parentheses. Time to extubation is presented as either mean ± SD¹⁷⁷ or median (first and third quartiles).¹⁷⁸

Sources: Adapted, with permission, from Brochard et al,¹⁷⁷ Esteban et al,¹⁷⁸ and Butler et al¹⁷⁹ for the risk-difference calculations.

In the study by Esteban et al,¹⁷⁸ 130 patients who met two of three weaning criteria and had failed a 2-hour T-piece trial were randomly assigned to one of four methods: SIMV (*n* = 29), PSV (*n* = 37), intermittent T-piece trials (two or more per day) (*n* = 37), and a once-daily T-piece trial (*n* = 31). The initial SIMV rate was set at half the frequency during ACV (average 10 breaths/min). When the patient tolerated it, SIMV was reduced twice a day by 2 to 4 breaths/min. Tracheal extubation was performed when the patient tolerated an SIMV rate of 5 breaths/min for 2 hours without signs of distress. With the PSV, the initial pressure was adjusted to achieve a frequency of 25 breaths/min or less (average pressure of 18 cm H₂O). According to patient tolerance, pressure was reduced at least twice a day by 2 to 4 cm H₂O. When the patient tolerated a

PSV level of 5 cm H₂O for 2 hours, tracheal extubation was performed. With intermittent T-piece trials, the patient breathed through a T-piece circuit or a continuous-flow CPAP of 5 cm H₂O or less at least twice a day. Trial duration was increased gradually, and when the patient tolerated a 2-hour trial, extubation was performed. Between the T-piece trials, ACV was applied. With the once-daily T-piece method, the patient breathed through a T-piece circuit, after which ACV was resumed for 24 hours. Trial duration was increased gradually. When the patient tolerated a 2-hour trial, extubation was performed. Weaning failure was defined as the need for reintubation within 48 hours after extubation or the inability to extubate the patient after 14 days. The median successful weaning duration was 5 days for SIMV, 4 days for PSV, 3 days for intermittent T-piece trials, and 3 days for the once-daily T-piece trials. The rate of successful weaning for the once-daily T-piece method was three times faster than with SIMV (rate ratio: 2.83) and two times faster than with PSV (rate ratio: 2.05). The rate of success for intermittent or once-daily T-piece trials did not differ significantly. The percentage of patients weaned successfully was 69%, 62%, 82%, and 71% for SIMV, PSV, intermittent T piece, and once-daily T piece, respectively.

These two large randomized studies showed conflicting results. In the study of Brochard et al,¹⁷⁷ PSV was superior to T piece and SIMV. In the study of Esteban et al,¹⁷⁸ T piece was superior to PSV and SIMV. Despite the subtle differences in methodology, both trials demonstrated that weaning time was the longest with SIMV.¹⁷⁹ Thus, both T piece and PSV were superior to SIMV (see Table 7-3).^{180,181}

In weaning preterm infants, Dimitriou et al¹⁸² compared pressure-limited SIMV with ACV in two separate randomized, controlled trials. With both SIMV and ACV ($n = 20$ each), inspiratory pressure was reduced in decrements of 2 cm H₂O until a defined target pressure tailored to the infant's body weight was reached. With SIMV, in addition to decreasing pressure, the SIMV rate was reduced in decrements of 5 breaths/min until a target rate of 20 breaths/min was reached in the first trial. In the second trial, the target was 5 breaths/min. The frequency of decrements in pressure or SIMV rate was not reported. When the infants tolerated the target pressure (for ACV) or target pressure and rate (for SIMV), ventilation was switched to CPAP for 1 hour before extubation. The end-expiratory pressure was maintained at 3 cm H₂O throughout the study. Weaning failure was defined as either the failure to achieve a reduction in ventilator support within 48 hours or requirement for reintubation within 48 hours of extubation. Reintubation was indicated when respiratory acidosis developed or frequent apneas or one major apnea occurred. In the first trial, there were no significant differences in the success rate with ACV or SIMV (70% vs. 75%) or the duration of successful weaning (median: 33 vs. 30 hours). In the second trial, differences between the weaning success rates were not significant, but the median weaning duration was significantly shorter with ACV (24 hours) than with SIMV (50 hours) ($P < 0.05$). A reduction in inspiratory pressure alone with ACV is favored in weaning preterm infants from mechanical ventilation than with SIMV, in which both inspiratory pressure and rate are reduced.

In weaning children ages 1 month to 4 years, Moraes et al³⁴ compared pressure-limited IMV ($n = 35$) and SIMV plus pressure support ($n = 35$). Weaning began when peak inspiratory pressure was less than 25 cm H₂O and $FiO_2 \leq 0.6$ (Time 0). Respiratory rate was reduced gradually by 3 to 5 breaths/min to 10 breaths/min. Then, PEEP was gradually reduced to 7 cm H₂O. These ventilator settings were maintained for 12 to 24 hours. Extubation readiness was assessed daily for 2 hours while on $FiO_2 \leq 0.5$ with oxygen saturation as measured using pulse oximetry (SpO_2) $\geq 95\%$, and PEEP 5 cm H₂O; in patients on SIMV plus pressure support, the pressure support level was tailored to the size of the endotracheal tube (pressure support of 10, 8, and 5 cm H₂O for endotracheal tube size of 3.0 to 3.5; 4.0 to 4.5; and >5.0 , respectively).¹⁸³ The average duration of weaning to extubation was 1 day (range: 1 to 6 days) for both groups. The frequency of

extubation failure of 5.7% (because of upper respiratory distress) was also similar in both groups. Unlike adults^{177,178} or preterm infants,¹⁸² the ventilator mode has no effects on the outcome of discontinuation from mechanical ventilation in children.³⁴

Intermittent Mandatory Ventilation and Mandatory Minute Volume, Adaptive-Support Ventilation

Mandatory minute ventilation (MMV) allows the patient to breathe spontaneously yet ensures that a preset minute ventilation is maintained should the patient's spontaneous ventilation decline below the set level.¹⁸⁴ MMV was developed to overcome certain ineffective features of IMV.¹⁸⁵ When the set mandatory IMV rate is less than required to achieve adequate ventilation, alveolar hypoventilation will ensue whenever a patient's total minute ventilation falls below a critical level. This drawback of IMV can be circumvented with MMV, which actuates a feedback control so that the ventilator provides pressurized breaths of a fixed volume to achieve a preset total minute ventilation.

Weaning with IMV and MMV was studied prospectively in forty patients recovering from acute respiratory failure caused by parenchymal lung injury and chronic airflow obstruction.¹⁸⁶ After meeting defined weaning criteria, the patients were randomized to IMV ($n = 18$) or MMV ($n = 22$). In the IMV group, IMV rate was decreased by 2 breaths/min at 3- to 4-hour intervals during the daytime only until the IMV rate was equal to zero. Weaning was considered complete after 4 hours of breathing on CPAP. In the MMV group, MMV was set at 75% of the total minute volume preceding the weaning trial; this was achieved by decreasing frequency while maintaining a V_T of 12 mL/kg as a reference value. Weaning was considered complete after 4 hours of independent spontaneous breathing. Weaning failure was defined as an inability to complete the trial or the need for ventilator support for the same underlying disease. Successful weaning was comparable: 86% for IMV and 89% for MMV. The weaning trial was longer in the IMV group (33 hours) than in the MMV group (4.75 hours).

In neonates with healthy lungs who were intubated for medical or surgical procedures, Guthrie et al¹⁸⁷ conducted a crossover design, short-term trial (2 hours) of MMV versus SIMV. Mandatory breaths with both MMV and SIMV were flow-limited, volume-cycled (V_T 4 to 6 mL/kg), whereas spontaneous breaths were augmented with pressure support. Both modes had comparable efficacy in carbon dioxide removal, yet with lower mean P_{aw} with MMV. Mean rate of the mandatory breaths was also significantly lower with MMV than with SIMV (4.1 vs. 24.2 breaths/min, respectively). The authors postulated that both the reduced rate of the mandatory breaths and low mean P_{aw} with MMV potentially reduced bronchopulmonary dysplasia complications associated with mechanical ventilation. Nevertheless, a prospective long-term follow-up is required. Moreover, SIMV as a weaning method has not been compared with MMV in neonates.

SIMV and ASV were compared as weaning modalities in a prospective, randomized study in post-cardiac surgery patients.¹⁸⁸ With both ASV ($n = 18$) and SIMV ($n = 16$), the patients underwent three ventilation phases. With ASV, in phase 1, the initial settings were the ideal body weight, the desired minute volume at the default value of 100 mL/kg of ideal body weight, and peak airway pressure of less than 25 cm H₂O. Adjustment of minute volume was dictated by a P_{aCO_2} of less than 38 or greater than 50 mm Hg. Phase 1 ended when there were no controlled breaths for 20 minutes. Phase 2 was a continuation of phase 1; it ended when pressure support was decreased to 10 cm H₂O (± 2 cm H₂O) and maintained for 20 minutes. The patient then entered into phase 3, where pressure support was set manually at 5 cm H₂O for 10 minutes. When the patient showed satisfactory tolerance, tracheal extubation was performed. The initial settings for phase 1 in the SIMV group consisted of a V_T of 8 mL/kg and an SIMV rate adjusted to achieve a P_{aCO_2} of between 38 and 50 mm Hg. The SIMV rate was then set at 12 breaths/min. When spontaneous breaths exceeded 6 breaths/min for 20

minutes, the patient was switched to PSV of 10 cm H₂O (phase 2). The patient was reassessed 20 minutes later for further reduction of PSV or returned to SIMV. If the patient tolerated it, PSV was reduced to 5 cm H₂O (phase 3), as in the ASV group. There was no difference in duration of tracheal intubation, and all patients except for two (one in each group) were extubated within 6 hours. In the ASV group, patients required fewer manipulations of ventilator settings and endured fewer high inspiratory pressure alarms. This study was performed in postoperative patients who had received mechanical ventilation for less than 24 hours before weaning attempts. Because mechanical ventilation duration before weaning influenced the weaning success rate,¹⁸⁹ the response of critically ill patients to the preceding weaning methods may be different.

Variation in Delivery among Ventilator Brands

No study has yet evaluated the response of various ventilators in the SIMV mode to patient flow demand or vice versa. As part of a study evaluating the response of muscle pressure generation to various unloading conditions with assisted ventilation, Mecklenburgh and Mapleson¹⁹⁰ evaluated the response of three ventilators—Hamilton Veolar, Engstrom Elvira, and Puritan Bennett 7200—in the SIMV mode in healthy subjects. V_T was set at 1.5 times the spontaneous V_T and the SIMV rate at 6 breaths/min. The flow waveform was set to “sine wave.” Muscle pressure was calculated using the equation of motion from instantaneous airway pressure, flow, and volume with the respiratory system’s known resistance and elastance.¹⁹¹ Amplitude of muscle pressure generation was similar across the three ventilators. Contraction time for mechanical breaths was shortest with the Engstrom Elvira at 1.03 seconds versus 1.38 seconds for the Hamilton Veolar and 1.37 seconds for the Puritan Bennett. For unassisted breaths, contraction time again was shortest with the Engstrom Elvira (1.33 vs. 1.58 seconds for the Hamilton Veolar and 1.70 seconds for the Puritan Bennett). Because V_T was set constant for all three ventilators, the short contraction time led to higher peak airway pressures with the Engstrom Elvira (9.4 vs. 3.9 cm H₂O for the Hamilton Veolar and 4.0 cm H₂O for the Puritan Bennett). Despite the set sine wave, the Engstrom Elvira flow waveform that accounted for the short contraction time was more of a ramp than a sine wave. The investigators concluded that differences in subject responses to different ventilators were related to flow or pressure waveforms and that different subjects may prefer different waveforms.

Adjustment at the Bedside and Troubleshooting

SIMV settings consist of the trigger sensitivity, V_T , flow, and the IMV rate for the flow-limit volume-cycled mandatory breaths. For pressure-limit time-cycled breaths, the ventilator settings include the trigger sensitivity, inspiratory pressure, inspiratory time, pressure attack rate, and IMV rate. If the dual hybrid breath-to-breath volume guarantee is applied, V_T is set instead of inspiratory pressure. With either flow-limited or pressure-limited mandatory breaths, pressure support, but not volume support, can be added to the spontaneous breaths to overcome circuit and endotracheal tube resistance, and unload inspiratory muscle work.⁴³ Monitoring of the patient and the ventilator output waveforms cannot be overemphasized.¹⁹² For example, palpable abdominal contractions suggest expiratory muscle recruitment and possible encroachment of mechanical inspiratory time into neural expiratory time.⁶⁴ Adjustment to reduce mechanical inspiratory time can be made by increasing flow rate (flow-limited breaths) or reducing inspiratory time (pressure-limited breaths). Despite the risk associated with tachypnea when mechanical inspiratory time is reduced, Laghi et al¹⁹³ demonstrated an increase in exhalation time and decrease in intrinsic PEEP, changes conducive to improved patient–ventilator interaction.

Important Unknowns

IMV has stood the test of time since its clinical application as a primary means of ventilator support in the early 1970s. To date, advanced technology enables most ventilators to be equipped with closed-loop ventilation, which allows full ventilator support with gradual support reduction. Unfortunately, few large, randomized, controlled trials have compared the efficacy of closed-loop ventilation with SIMV with or without pressure support in terms of mechanical ventilation duration, patient–ventilator interaction, sensation of dyspnea, and ventilator-associated complications.

Studies show that inspiratory muscle activity is of the same intensity during machine assistance as during the intervening spontaneous breaths,⁵⁷ and that SIMV prolongs weaning.^{177,178} Given that the diaphragm is activated with each breath, it is possible that SIMV protects the respiratory muscles from disuse atrophy, which occurs with CMV.^{82,84,85} Alternatively, the increased workload at low levels of machine assistance actually may cause overload^{194,195} and prolong mechanical ventilation duration or weaning time. Which of those two factors plays a role during a low assistance level of SIMV is unknown.

The Future

Several studies demonstrate a decline in the application of combined SIMV and pressure support in acute respiratory failure,^{10,196} except in less-severity-of-illness, postoperative, and trauma patients.¹² In patients with ARDS, the proportion of patients managed with SIMV has declined from 22% in 1996 to 3% in 2005.¹⁹⁶ With the availability of closed-loop ventilation with improved patient–ventilator synchrony,⁷² SIMV will likely be displaced. In contrast to use of SIMV in adults, the ability to employ simple ventilator settings with SIMV and the options of combining it with pressure support and of guaranteeing volume with use of pressure-limited mandatory breaths, ensures that SIMV will remain an important primary ventilator mode in critically ill pediatric patients.¹⁴

Summary and Conclusions

Few ventilator modes have advanced our understanding of the mechanisms that underlie patient–ventilator interaction as has SIMV. As primary means of ventilator mode in less critically ill and postoperative patients in adults and critically ill pediatrics, SIMV remains one of the most widely used modes of ventilation, as does ACV. No mortality difference has been observed between SIMV and ACV. To date, no large randomized study has compared SIMV with more technologically advanced modes as primary methods of ventilation. As a weaning technique, SIMV has been shown to be inferior to T piece and PSV.

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